

Integrate Zscaler with Identity Providers (IdPs) such as Azure AD

Title of the project :-

Integrate Zscaler with Identity Providers (IdPs) such as Azure AD
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INTRODUCTION:

In an era where the digital landscape is constantly evolving, organizations face an ever-increasing threat from cyberattacks. The need for swift and effective incident response has become paramount to safeguard sensitive data, protect digital assets, and maintain the trust of stakeholders. In response to this growing challenge, we present a visionary project: the development of an AI-driven incident response platform that empowers cybersecurity teams with the tools to automate and enhance incident triage and response tasks.

In response to the evolving threat landscape and the dissolution of the traditional network perimeter, modern enterprises are rapidly shifting away from legacy VPN-centric remote access models. These conventional approaches often grant excessive network-level access, creating a significant attack surface and increasing the risk of lateral threat movement. To address these challenges, this project centers on the implementation of a Zero Trust security architecture, fundamentally operating on the principle of "never trust, always verify." By moving beyond location-based access, we aim to establish a security posture where trust is no longer implicit, and access is determined by granular context rather than network connectivity.

The core of this initiative involves the deployment of Zscaler Private Access (ZPA), a cloud-delivered Zero Trust Access solution designed to create identity-aware, application-specific connectivity. Unlike traditional VPNs that place users on the internal network, ZPA leverages a "service-initiated" model to establish direct, one-to-one connections between authorized users and specific applications, completely hiding the corporate network and IP address space from the internet. This ensures that users connect exclusively to the applications they are permitted to access, based on continuous identity verification and context, rendering the applications invisible to unauthorized users and preventing common attack vectors such as port scanning and distributed denial-of-service (DDoS) attacks.

Ultimately, this project will enforce a least-privilege access model, ensuring users and devices are granted the minimum required access necessary to perform their functions. By decoupling application access from network access, we will eliminate the risk of lateral movement across the network, as users have no visibility or connectivity to other hosts. This strategic move not only significantly enhances security by reducing the attack surface but also simplifies the user experience and reduces operational overhead associated with legacy remote access infrastructure, positioning the organization at the forefront of secure and efficient access management.

ABSTRACT:

The proliferation of hybrid work and cloud-based resources has rendered traditional perimeter-based security models, such as VPNs, obsolete and increasingly dangerous. These legacy solutions often provide overly broad network access, exposing organizations to significant risks including lateral threat movement and an expanded attack surface. This project addresses these vulnerabilities by implementing a Zero Trust Architecture (ZTA) utilizing Zscaler Private Access (ZPA). The primary objective is to transition from network-centric to identity-centric access, thereby enforcing the principle of least privilege.

This project details the integration of the Zscaler Internet security platform with leading Identity Providers (IdPs), including Microsoft Entra ID (Azure AD), Okta, and on-premises Active Directory (AD). The primary objective is to establish a unified identity framework that enables seamless Single Sign-On (SSO) and centralized authorization for all Zscaler services. By consolidating authentication into trusted corporate directories, this initiative eliminates the need for disparate credentials and provides a foundational element for a modern Zero Trust security architecture. By leveraging industry-standard protocols such as SAML, OIDC, and SCIM, this integration ensures that user authentication is delegated to trusted corporate directories while enforcing consistent security policies, including Multi-Factor Authentication (MFA).

This approach not only enhances the user experience by reducing password fatigue but also strengthens the organization's security posture by ensuring that every access request is verified by the identity provider before reaching Zscaler's security enforcement points. Furthermore, the implementation of automated user provisioning via SCIM streamlines lifecycle management, reducing administrative overhead and enhancing security posture. The outcome is a robust, identity-centric security model that aligns with Zero Trust principles, ensuring that only authenticated and authorized users gain secure access to internet and cloud applications from any location. This integration effectively bridges the gap between identity governance and network security, creating a dynamic and responsive access control environment.

Use Case Scenarios

Scenario 1: Finance Department Access to ERP System

A senior accountant in the finance department needs to access the corporate Enterprise Resource Planning (ERP) system to process monthly payroll. The user connects from a remote location using their company-managed laptop. Upon launching the ZPA Client Connector, they authenticate via Azure Active Directory with multi-factor authentication (MFA). ZPA evaluates the user's identity, group membership (Finance_ERP_Users), device posture (antivirus active, disk encryption enabled), and connection context. The policy engine determines that the user is authorized and brokers a secure, direct micro-tunnel to the ERP application only. The user can interact with the ERP interface, but has no visibility or network-level access to any other internal systems, such as development servers or the HR database.

Output:

- User authenticated and authorized based on identity and group membership
- Direct micro-tunnel established specifically to the ERP application
- All other internal applications and network segments remain invisible and inaccessible

Scenario 2: Contractor Access to Specific Project Repository

An external software consultant is hired for a three-month project and requires access only to a specific Git repository hosted on an internal server. The contractor is onboarded as a guest user in the identity provider and assigned to the "Project_X_Contractors" group. When the contractor connects using the ZPA client, the system verifies their identity and checks the device for basic security compliance (OS patch level, firewall enabled). ZPA enforces a policy that grants access exclusively to the Git repository application (port 22/SSH or port 443/HTTPS) during business hours (9 AM to 5 PM). The contractor can clone, push, and pull code, but cannot discover or connect to any other internal resources, including the corporate intranet or file servers.

Output:

- Time-bound and role-specific access enforced
- Contractor restricted to a single application (Git repository)
- Lateral movement to adjacent systems completely blocked

Scenario 3: Executive Access to Confidential Board Documents

A company executive needs to access a confidential folder containing board meeting materials hosted on an internal file server. The executive connects from a tablet device while traveling internationally. ZPA integrates with the mobile device management (MDM) solution to verify that the tablet is compliant (not jailbroken, passcode enabled, encryption active). Upon successful authentication, ZPA evaluates the policy: "If user is in 'Executive_Board' group AND device is MDM-compliant AND connection is from a trusted country (allowlist), THEN grant access to 'Board_Documents' application." The executive is connected directly to the specific file share, and all data in transit is encrypted via TLS. Any attempt to access other shares or network resources is automatically blocked and logged.

Output:

- Context-aware access based on user role, device posture, and geographic location
- Granular access to a specific file share (Board_Documents)
- Comprehensive logging of access events for audit and compliance

Scenario 4: IT Administrator Access to Server Management Interfaces

A systems administrator requires access to management interfaces (SSH for Linux servers and RDP for Windows servers) to perform routine maintenance. Instead of granting VPN access to the entire management network, ZPA is configured to treat each server or management interface as a separate application segment. The administrator authenticates and is presented only with the specific servers they are authorized to manage based on their IT subgroup (e.g., "Linux_Admins" see only Linux servers; "Windows_Admins" see only Windows servers). When the admin connects via SSH to a Linux server, ZPA creates a direct micro-tunnel to that specific IP and port combination. The admin cannot scan the subnet, access other servers, or initiate connections to non-authorized ports.

Output:

- Micro-segmentation at the individual server/interface level
- Role-based access controlling visibility of specific management consoles
- Elimination of network-level reconnaissance capabilities for admins

Scenario 5: Mergers and Acquisitions Integration

A company acquires a smaller firm and needs to provide the acquired company's HR team with temporary access to the parent company's benefits enrollment application. The acquired company's users are added as external collaborators in the identity provider with limited attributes. ZPA policies are created that specifically grant the "Acquired_HR" group access only to the "Benefits_App" (a web application hosted internally). The acquired users install the ZPA client, authenticate, and are connected directly to the benefits application. Because ZPA provides access without network connectivity, the parent company's entire internal network remains completely hidden and protected. The acquired users cannot ping, scan, or access any other corporate resource. After six months, when the integration is complete and users are migrated to the parent identity system, the temporary policies are revoked.

Output:

- Secure, temporary access provided without network exposure
- Complete network invisibility protecting the parent company's infrastructure
- Simplified deprovisioning through policy revocation rather than complex network changes

Title:

Integrate Zscaler with Identity Providers (IdPs) such as Azure AD

Overview:

In response to the evolving threat landscape and the dissolution of the traditional network perimeter, modern enterprises are rapidly shifting away from legacy VPN-centric remote access models. These conventional approaches often grant excessive network-level access, creating a significant attack surface and increasing the risk of lateral threat movement. To address these challenges, this project centers on the implementation of a Zero Trust security architecture, fundamentally operating on the principle of "never trust, always verify." By moving beyond location-based access, we aim to establish a security posture where trust is no longer implicit, and access is determined by granular context rather than network connectivity.

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List of Vulnerability:

S. No.	Vulnerability Name	CWE No.
1.	Cross-site scripting (stored)	79
2.	Broken Access Control	285
3.	SQL Injection	89
4.	Broken Authentication	285
5.	Insecure Direct Object Reference	639
6.	Security Misconfiguration	732
7.	Cross-Site Request Forgery	352
8.	Cleartext Transmission of Sensitive Information	319
9.	ClickJacking	1021
10.	webserver allows password auto-completion	310
11.	Cross-site scripting (DOM-based XSS)	79
12.	JQuery 1.2 < 3.5.0 Multiple XSS	79
13.	Information Exposure (Web Server HTTP Header Internal IP Disclosure)	200
14.	Information Exposure (SSL (Multiple Issues))	200
15.	Information Exposure (HTTP (Multiple Issues))	200
16.	OS Identification Failed	200

Report:

1. Vulnerability name: Cross site scripting (stored)

CWE: 79

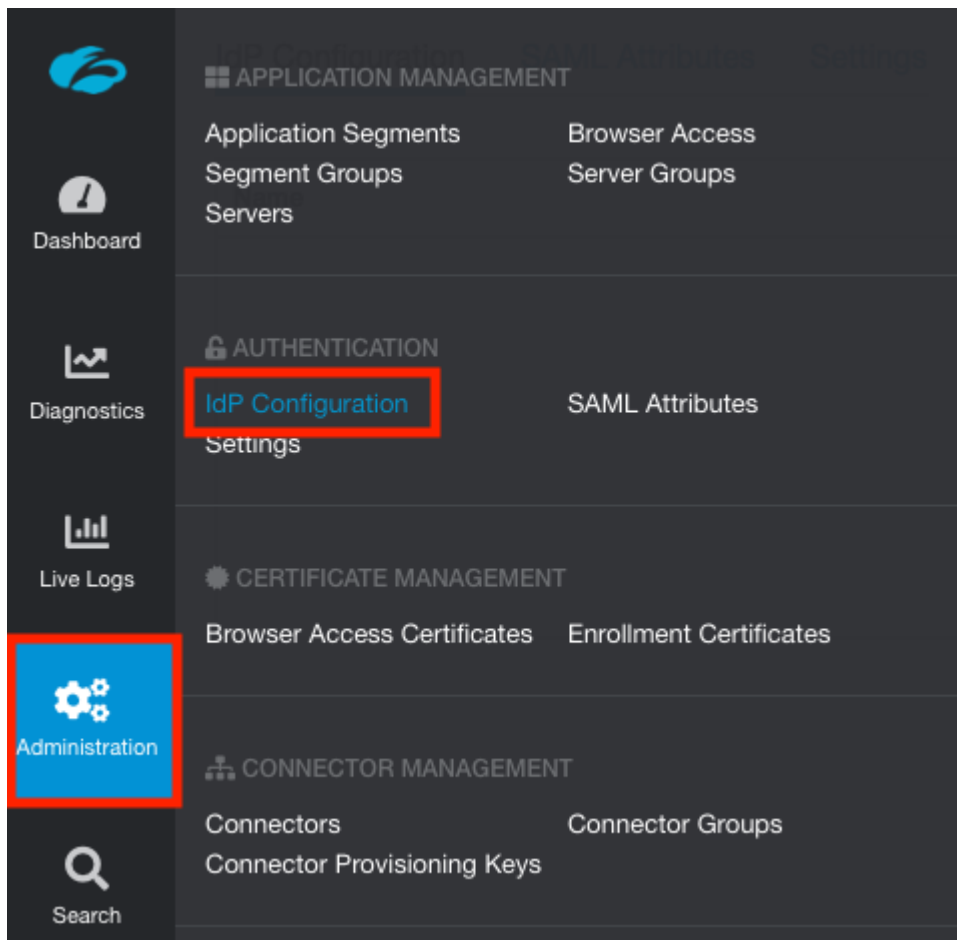
OWASP category: A03:2021 -Injections

Description: It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser.

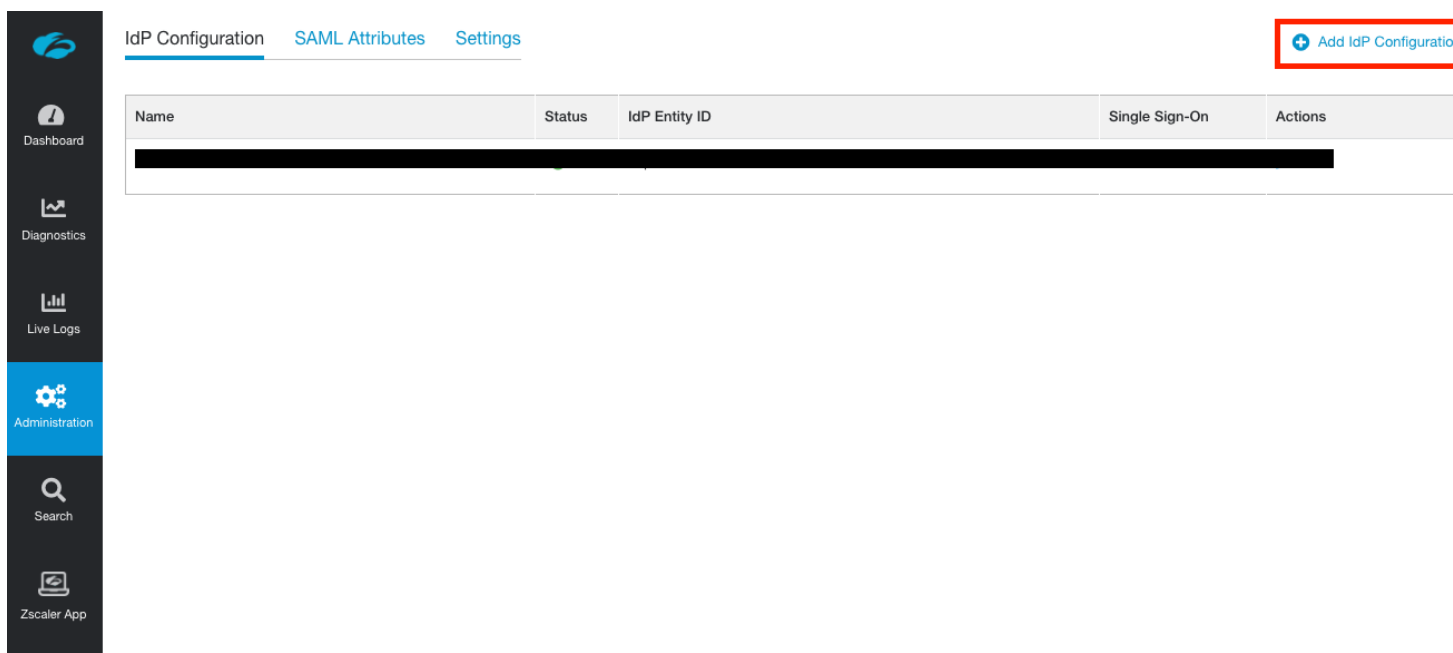
Business Impact: The application stores dangerous data in a database, message forum, visitor log, or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area that is displayed to either many users or particularly interesting users. Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data belonging to the user.

Add a New Identity Provider in ZPA

Log into the [ZPA admin portal](#) and go to **Administration > IdP Configuration**.



Click **Add IdP Configuration** at the top-right of the screen.



Under the **IdP Information** section:

1. Enter a name for the IdP configuration. ZPA supports multiple IdPs, hence a name is required to differentiate them.
2. Ensure **User** is selected for **Single Sign-On**.

3. Select the domain(s) you wish to associate with the IdP configuration. Zscaler support must add any additional domains to your admin portal first before you can use them.

Add IdP Configuration

1 IdP Information

2 SP Metadata

3 Create IdP

Name

Azure AD - Oblivion Industries

Single Sign-On

Admin

☒ User

Domains

× oblivion.industries

Next

Quit

Click **Next** to continue.

The next tab shows info you'll for the configuration on the Azure AD side. **Download the Service Provider Metadata file** (this will download as a file called **sp_metadata.xml**) as we'll use this in the next section.

Add IdP Configuration



1 IdP Information

2 SP Metadata

3 Create IdP

Configure the Service Provider information in your IdP

SERVICE PROVIDER SAML METADATA FOR USER SSO

Service Provider Metadata
[Download Metadata](#)

Service Provider Certificate
[Download Certificate](#)

Service Provider URL

Service Provider Entity ID

Next

Pause

Click **Pause** (this will suspend the IdP config temporarily) and continue to the next section.

Add ZPA as an Enterprise Application in Azure AD

From the [Azure Portal](#), open the side hamburger menu and click **Azure Active Directory**. You can also search for it using the search bar at the top.

Under **Manage** in the side menu, click **Enterprise applications**.

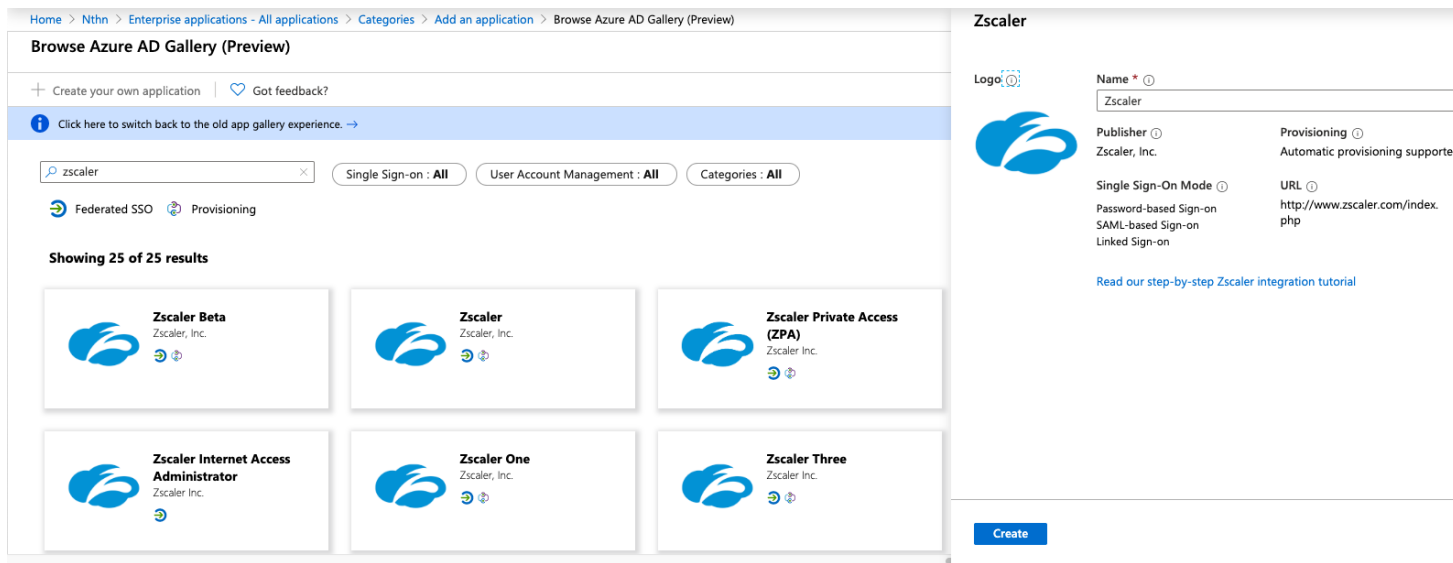
The screenshot shows the Microsoft Azure portal interface. The left sidebar contains a 'Manage' section with various options. 'Enterprise applications' is highlighted with a red box. The main content area shows the 'Oblivion Industries' overview, including tenant ID, role, and a line graph for sign-ins.

Next, under the **All Applications** menu, click **New Application**.

The screenshot shows the Microsoft Azure portal interface for 'Enterprise applications | All applications'. The 'New application' button is highlighted with a red box. Below it, a banner encourages trying the new Enterprise Apps search preview. The main content area displays a table of applications with columns for Name, Homepage URL, Object ID, and Application ID.

Name	Homepage URL	Object ID	Application ID
dxprovisioning-worker-mfa			
Microsoft Teams			
MOD Demo Platform UnifiedApiConsumer			
Office 365 Exchange Online	http://office.microsoft.com/outlook/		
Office 365 Management APIs			
Office 365 SharePoint Online	http://office.microsoft.com/sharepoint/		
Office 365 Yammer	https://products.office.com/yammer/		
Outlook Groups			
Power BI Service			
Skype for Business Online			

In the search box to add a new application, type "**Zscaler**".



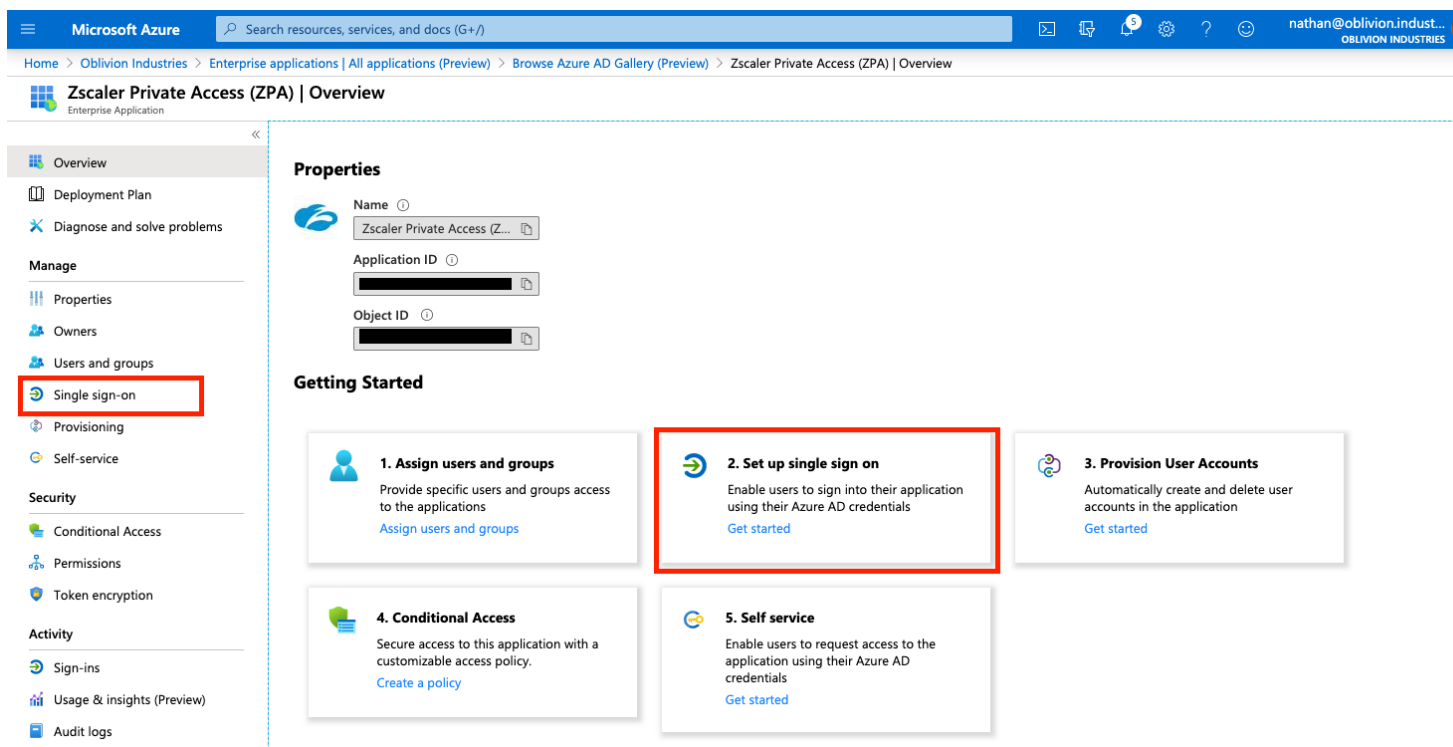
Select the **Zscaler Private Access (ZPA)** application and click **Create** (or **Add** depending on whether you are using the old or new enterprise app gallery).

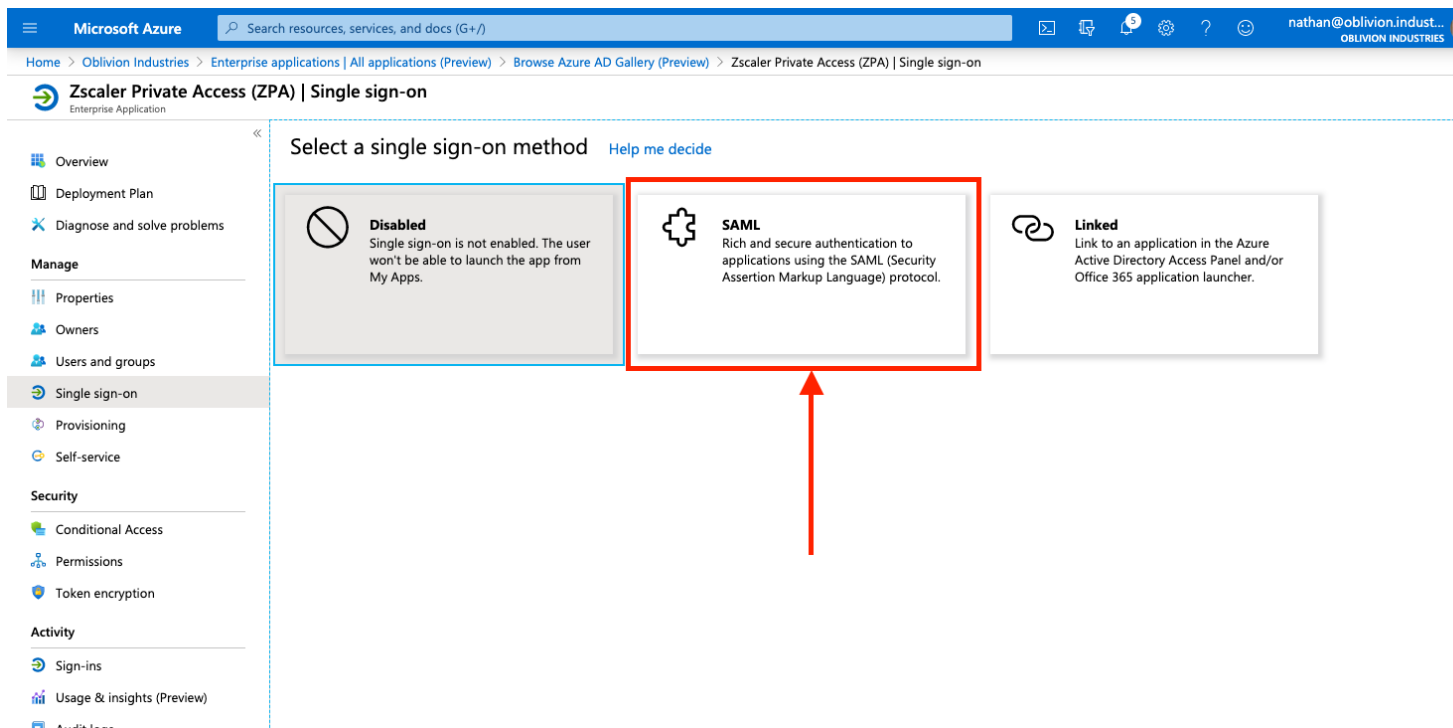
Configure the ZPA Enterprise Application

You will now be on the administration page for the ZPA enterprise application you added above (if not, you can get to this page via: Azure Active Directory > Enterprise Applications > All Applications, then click on the ZPA app you added above).

1 - Configure SAML-based Single Sign-on

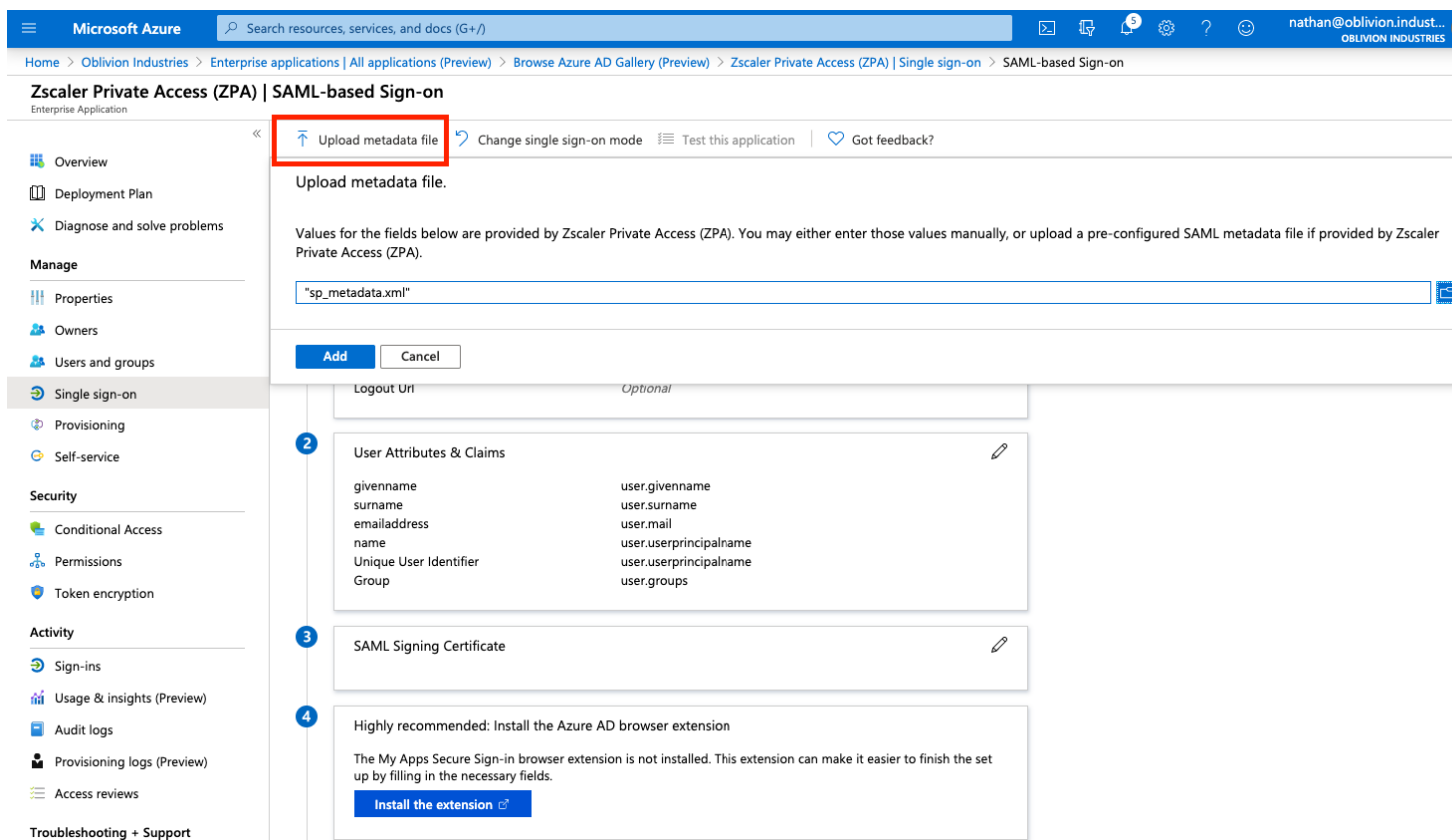
Under **Manage** in the side menu, click **Single sign-on**, and select **SAML** when prompted. This will place you on the SAML configuration page.





2 - Import the Zscaler SP Metadata File

Click **Upload metadata file** at the top, and select the **sp_metadata.xml** file you downloaded from the ZPA admin portal earlier. This will open a panel and pre-populate some of the required info.



The Sign on URL field will be empty and is required. Copy the example URL, replacing the EXAMPLE at the end with your domain that is being associated with this Azure AD config.

For example, for my domain of oblivion.industries, my Sign on URL would look like:

<https://samlsp.private.zscaler.com/auth/login?domain=oblivion.industries>

Microsoft Azure | Search resources, services, and docs (G+/)

nathan@oblivion.industries

Home > Oblivion Industries > Enterprise applications | All applications (Preview) > Browse Azure AD Gal

Zscaler Private Access (ZPA) | SAML-based Sign-on

Enterprise Application

Overview

Deployment Plan

Diagnose and solve problems

Manage

- Properties
- Owners
- Users and groups
- Single sign-on
- Provisioning
- Self-service

Security

- Conditional Access
- Permissions
- Token encryption

Activity

- Sign-ins
- Usage & insights (Preview)
- Audit logs
- Provisioning logs (Preview)
- Access reviews

Troubleshooting + Support

- Virtual assistant (Preview)
- New support request

Set up Single Sign-On with SAML

Read the [configuration guide](#) for help integrating Zscaler

- Basic SAML Configuration**
 - Identifier (Entity ID) * ⓘ
The default identifier will be the audience of the SAML response for IDP-initiated SSO
Default
https://samlsp.private.zscaler.com/auth/metadata/144133171103924285
 - Reply URL (Assertion Consumer Service URL) * ⓘ
The default reply URL will be the destination in the SAML response for IDP-initiated SSO
Default
https://samlsp.private.zscaler.com/auth/144133171103924285/sso
 - Sign on URL * ⓘ
Patterns: https://samlsp.private.zscaler.com/auth/login?domain=oblivion.industries
 - Relay State ⓘ
Enter a relay state
 - Logout Url ⓘ
Enter a logout url
- User Attributes & Claims
- SAML Signing Certificate
- Highly recommended: Install the Azure AD browser extension
- Set up Zscaler Private Access (ZPA)

The Relay State and Logout Url fields should be left blank.

Click **Save** at the top when you are done, and X to close the panel.

3 - Create a SAML Signing Certificate

Next to the 3rd step, **SAML Signing Certificate** in the SAML configuration page, click the edit/pencil icon.

Microsoft Azure Search resources, services, and docs (G+)

Home > Oblivion Industries > Enterprise applications | All applications (Preview) > Browse Azure AD Gallery (Preview) > Zscaler Private Access (ZPA) | Single sign-on > SAML-based Sign-on

Zscaler Private Access (ZPA) | SAML-based Sign-on

Enterprise Application

Upload metadata file Change single sign-on mode Test this application Got feedback?

Set up Single Sign-On with SAML

Read the [configuration guide](#) for help integrating Zscaler Private Access (ZPA).

- Basic SAML Configuration**
 - Identifier (Entity ID): `https://samlsp.private.zscaler.com/auth/metadata/144133171103924285`
 - Reply URL (Assertion Consumer Service URL): `https://samlsp.private.zscaler.com/auth/144133171103924285/sso`
 - Sign on URL: `https://samlsp.private.zscaler.com/auth/login?domain=oblivion.industries`
 - Relay State: *Optional*
 - Logout Url: *Optional*
- User Attributes & Claims**
 - givenname: `user.givenname`
 - surname: `user.surname`
 - emailaddress: `user.mail`
 - name: `user.userprincipalname`
 - Unique User Identifier: `user.userprincipalname`
 - Group: `user.groups`
- SAML Signing Certificate**
- Highly recommended: Install the Azure AD browser extension

In the panel that appears, make sure the **Signing Algorithm** is **SHA-256** and **Signing Option** is set to **Sign SAML assertion**.

Click **New Certificate** then **Save**. Open the menu by clicking the three dots to the right of the certificate, and download the BOTH the **Base64** certificate and the **Federated Certificate XML**. You should only need the latter, but it can sometimes be helpful to have both.

SAML Signing Certificate

Manage the certificate used by Azure AD to sign SAML tokens issued to your app

Save + New Certificate Import Certificate

Status	Expiration Date	Thumbprint	
n/a	05/10/2023	Will be displayed on save	...

Signing Option: Sign SAML assertion

Signing Algorithm: SHA-256

Notification Email Addresses: nathan@oblivion.industries

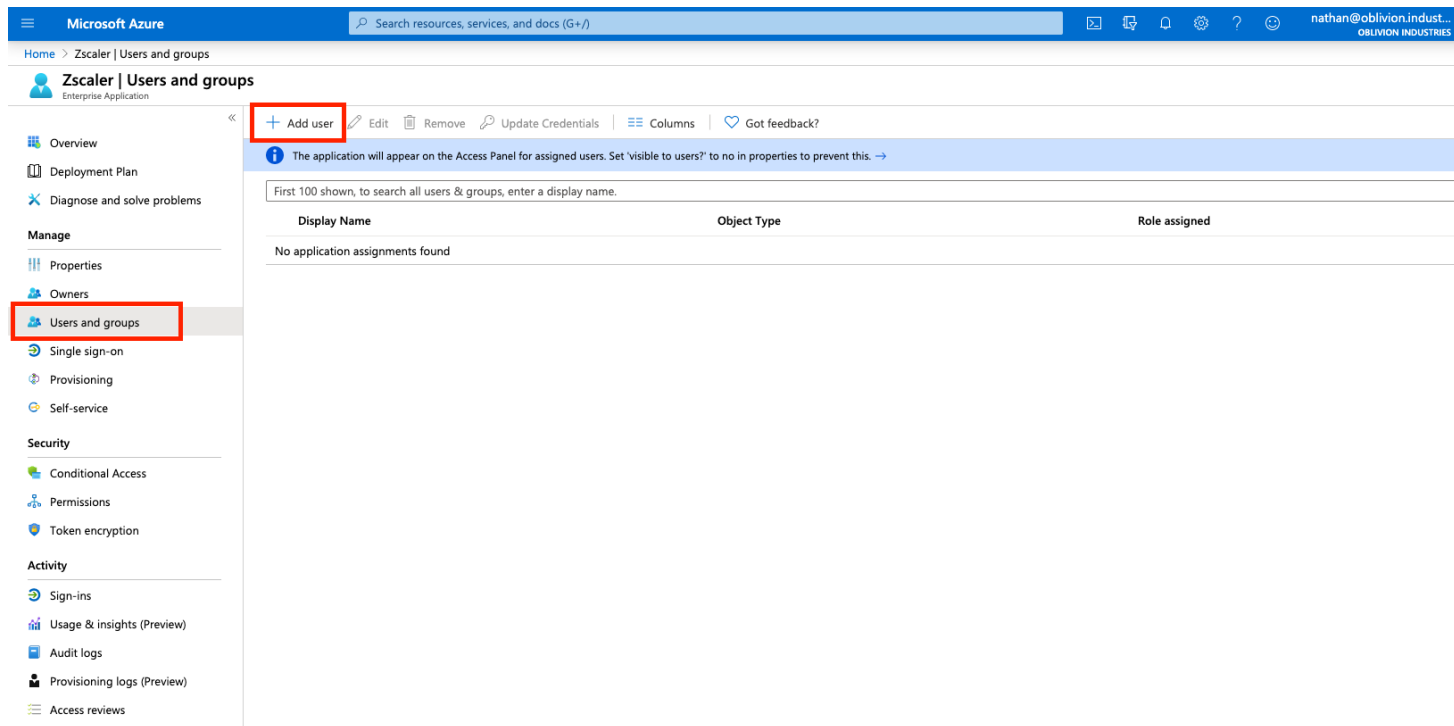
- Make certificate active
- Base64 certificate download
- PEM certificate download
- Raw certificate download
- Download federated certificate
- Delete Certificate

Close the panel by clicking X once you've downloaded the certificate and metadata files.

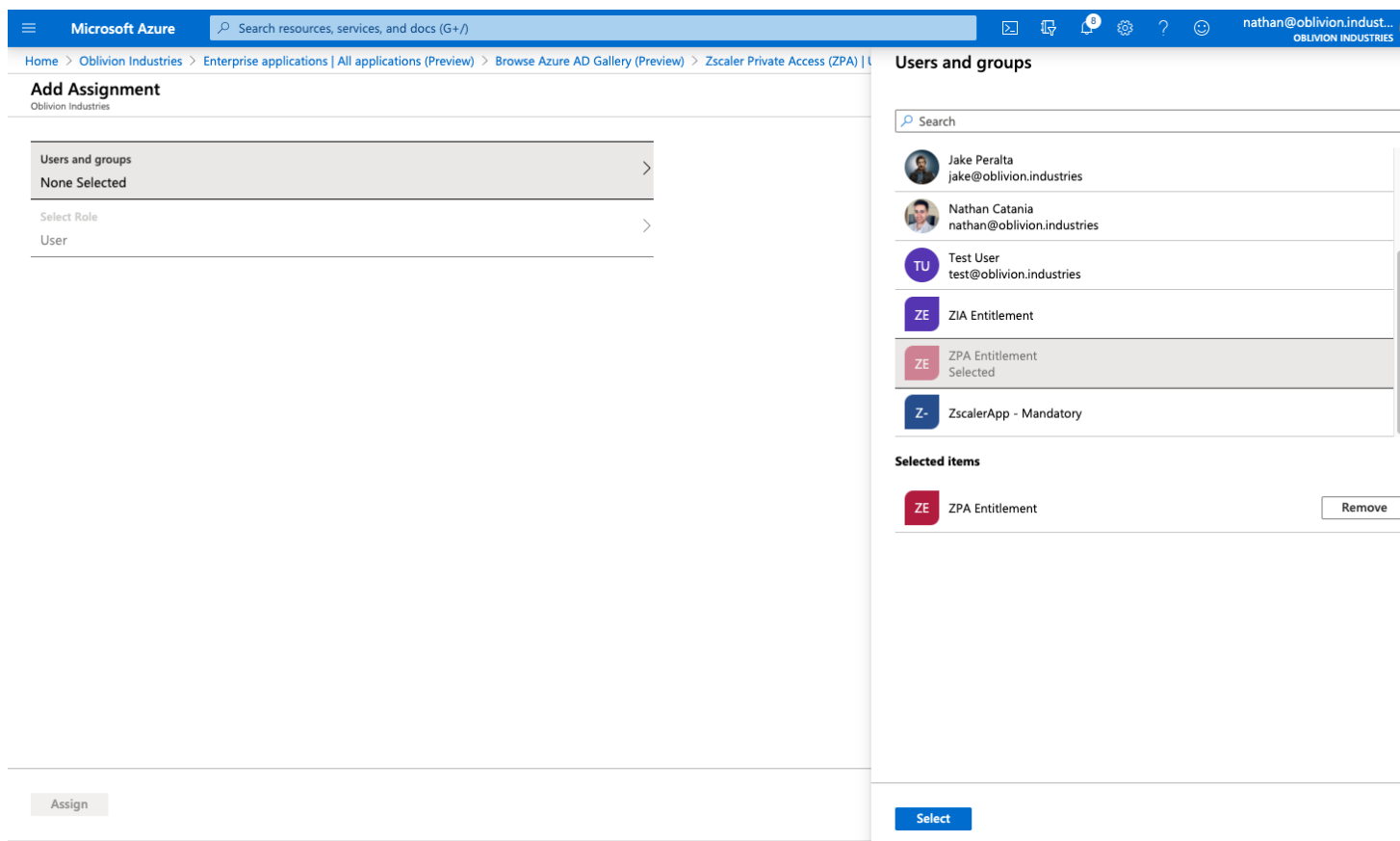
Assign Users & Groups to the Enterprise Application

We now need to assign what users are authorized to use the ZPA Enterprise Application. Only the users or groups specified will be able to sign into ZPA via Microsoft SSO. If you skip this section, your user's won't be able to sign in, and you won't see user data populated in ZIA for you to configure policy around.

Select **Users and groups** from the side menu of the enterprise application, then click **Add user**.



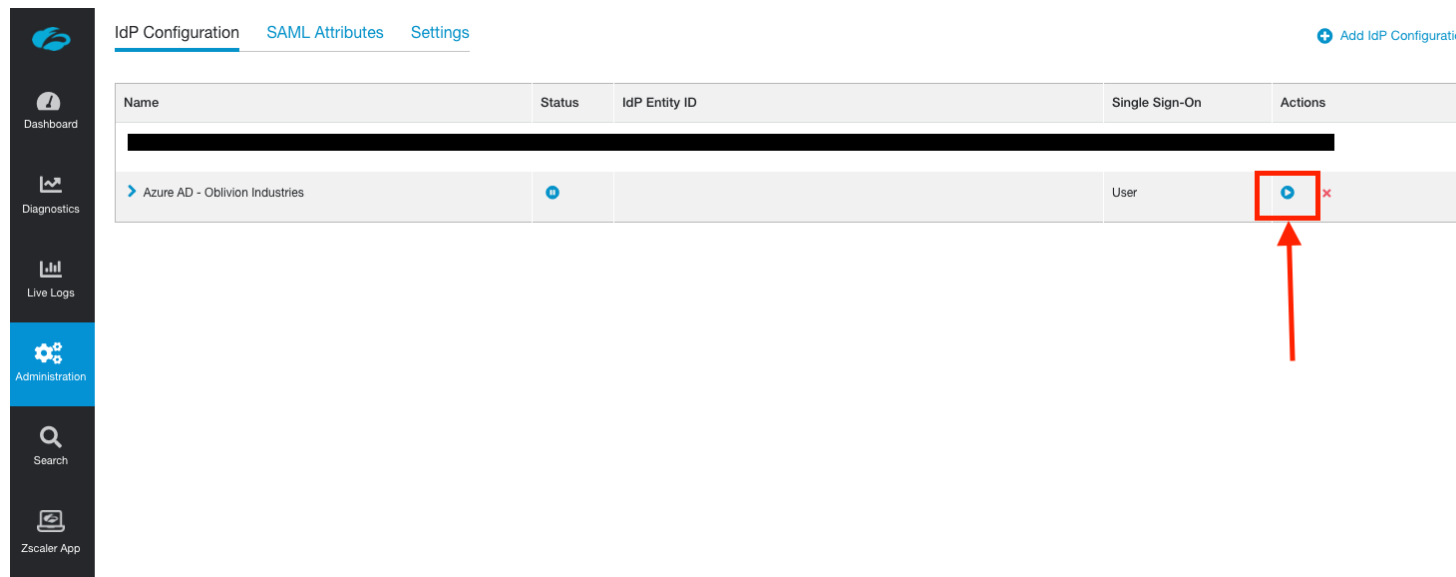
Select the Users and Groups to be both synced with and provisioned access to ZPA. If you select a group, all members of that group will receive ZPA access.



When you are finished, click **Assign**.

Finish Configuring the IdP in ZPA

Return to the ZPA admin portal and click the **resume** button next to the paused IdP configuration (Administration > IdP Configuration).



The screenshot shows the ZPA admin portal interface. On the left is a dark sidebar with navigation icons for Dashboard, Diagnostics, Live Logs, Administration (highlighted in blue), Search, and Zscaler App. The main content area has tabs for 'IdP Configuration', 'SAML Attributes', and 'Settings'. Below the tabs is a table with columns: Name, Status, IdP Entity ID, Single Sign-On, and Actions. The table contains one entry: 'Azure AD - Oblivion Industries' with a status of 'Paused' (indicated by a blue circle with a white 'P'). The 'Actions' column for this entry contains a blue circular button with a white play icon, which is highlighted by a red square and a red arrow pointing upwards. To the right of this button is a red 'x' icon. In the top right corner of the main area, there is a link that says '+ Add IdP Configuration'.

Name	Status	IdP Entity ID	Single Sign-On	Actions
Azure AD - Oblivion Industries	Paused		User	 

For the **IdP Metadata File**, click to **upload the Federated Certificate XML** you download from Azure AD above. *This will automatically populate all settings for you; including the certificate.*

If for some reason the certificate does not populate, you can upload the Base64 certificate you downloaded.

1 IdP Information

2 SP Metadata

3 Create IdP

Name

Azure AD - Oblivion Industries

Authentication Domains

✕ oblivion.industries

SAML ATTRIBUTES

IdP Metadata File

Zscaler Private Access (ZPA) (3).xml

Change

Remove

IdP Certificate

Upload the Certificate File...

Select File



-----END CERTIFICATE-----

Single Sign-On URL

https://login.microsoftonline.com/[REDACTED]/saml2

IdP Entity ID

Save

Quit

Scroll to the bottom of the window and change **SCIM Sync** to **Enabled**.

Note down the **SCIM Service Provider Endpoint** URL, and generate a new **Bearer Token**. Note down the Bearer Token as well. You will need both of these in the next step.

Add IdP Configuration

-----END CERTIFICATE-----

Single Sign-On URL
[https://login.microsoftonline.com/\[REDACTED\]/saml2](https://login.microsoftonline.com/[REDACTED]/saml2)

IdP Entity ID
[https://sts.windows.net/\[REDACTED\]/](https://sts.windows.net/[REDACTED]/)

Status
☒ Enabled ☐ Disabled

ZPA (SP) SAML Request
☒ Signed ☐ Unsigned

HTTP-Redirect
☒ Enabled ☐ Disabled

SCIM

SCIM Sync
☒ Enabled ☐ Disabled

SCIM Service Provider Endpoint
[https://scim1.private.zscaler.com/scim/1/\[REDACTED\]/v2](https://scim1.private.zscaler.com/scim/1/[REDACTED]/v2)

Bearer Token
[REDACTED]

[Generate New Token](#)

Save

Quit

Click **Save** to finish the configuration on the ZPA side. Unlike ZIA, you do not need to activate your changes for them to take effect.

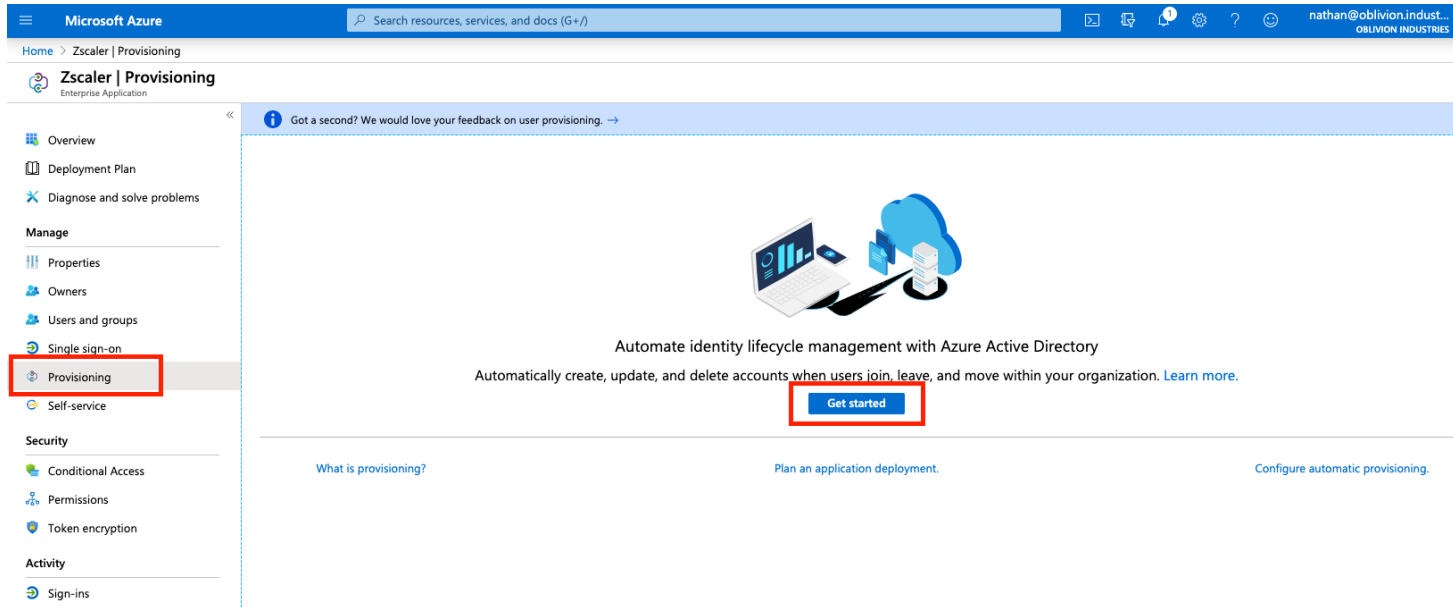
Configure SCIM Provisioning in Azure AD

So far we've configured a way to authenticate our users, but we still need to configure a method to provision & de-provision them in the ZPA database.

You can learn more about SCIM Provisioning [here](#), but in short, it involves the IdP regularly syncing user information with Zscaler via an API. With Azure AD, this sync happens every 40 minutes. Using SCIM is best practice.

1 - Configure SCIM Provisioning in Azure AD

Return to the ZPA Enterprise Application in Azure AD. This time select **Provisioning** from the side menu and select **Get Started** if prompted.



Change the **Provisioning Mode** to **Automatic**, and fill in the following fields:

- For **Tenant URL**, copy and paste the **Base URL** from above.
- For **Secret Token**, copy and paste the **Bearer Token** from above.

Click **Test Connection** to verify. If you receive an error about "invalid credentials", make sure you saved and activated your change in the step above.

Microsoft Azure Search resources, services, and docs (G+/)

Home > Oblivion Industries > Enterprise applications | All applications (Preview) > Browse Azure AD Gallery (Preview) > Zscaler Private Access (ZPA) | Provisioning > Provisioning

Provisioning

Save Discard

i This provisioning connector is in preview. Please click here to provide us feedback.

Provisioning Mode Automatic

Use Azure AD to manage the creation and synchronization of user accounts in Zscaler Private Access (ZPA) based on user and group assignment.

Admin Credentials

Admin Credentials

Azure AD needs the following information to connect to Zscaler Private Access (ZPA)'s API and synchronize user data.

Tenant URL *

Secret Token *

Test Connection

Notification Email

☐ Send an email notification when a failure occurs

Mappings

Settings

Start and stop provisioning to Zscaler Private Access (ZPA), and view provisioning status.

Provisioning Status ☒ On ☐ Off

Scope

☐ Clear current state and restart synchronization

Click **Save** to continue.

Once your changes have been saved, change **Provisioning Status** to **On**.

Save your changes again and the Scope field should appear.

If you receive an error, or the scope field is not appearing:

- Back out of the Provisioning menu and go into it again. This should fix the error and cause the scope menu to appear.
- Ensure you have assigned users and/or groups (containing users) to the enterprise application first.

For the **Scope**:

Scope	Description
Sync only assigned users and groups	Only users and groups explicitly assigned access to this enterprise application (under Users and groups in the side menu) will be provisioned via SCIM to ZPA.

Scope	Description
Sync all users and groups	All users and groups defined in Azure AD are synced with ZPA; irrespective of what was defined under Users and Groups in the side menu.

Set your scope accordingly and click **Save** again. Don't forget to do this or your user information will not sync with ZPA!

3 - Synchronize

Your users will sync to ZPA on the next API cycle (every 40 minutes). As this is the first time, you can manually force a sync by clicking the option **Clear current state and restart synchronization** (and then **Save**).

4 - Verify

Once your users and groups have synchronised with ZPA, they'll be visible in the ZPA admin portal under **Administration > SCIM Users** and **Administration > SCIM Groups**.

The screenshot shows the ZPA admin portal interface. On the left is a sidebar with navigation icons for Dashboard, Diagnostics, Live Logs, Administration (highlighted), Search, and Zscaler App. The main content area is titled 'SCIM Attributes' and has tabs for 'SCIM Users' (selected) and 'SCIM Groups'. Below the tabs are filters: 'SCIM ENABLED IDENTITY PROVIDERS' set to 'Azure AD - Oblivion Industries', 'SCIM USER NAME' with a search bar, 'IDP SCIM USER ID' with a search bar, and 'SCIM GROUP NAME' set to 'None'. There is also a 'TIME RANGE' filter set to 'Nov 10, 2019 AEST - May 10, 2020 AEST' and an 'Update' button. Below these filters is a table of users. The table has columns: 'Last Updated' (with a refresh icon), 'UTC', 'Internal User ID', 'SCIM User Name', and 'IdP SCIM User ID'. Two users are listed: 'test@oblivion.industries' with ID 390883 and 'nathan@oblivion.industries' with ID 390882. The table is highlighted with a red border.

Last Updated	UTC	Internal User ID	SCIM User Name	IdP SCIM User ID
May 10th, 18:58 (AEST)		390883	test@oblivion.industries	
May 10th, 18:58 (AEST)		390882	nathan@oblivion.industries	

Import SAML Attributes

In order to create policies in ZPA, we need to import the fields we can use (called SAML attributes) from the IdP. This also allows us to test whether user authentication is working correctly.

IMPORTANT: I strongly recommend do the next step in an Incognito or Private tab. Failing that, completely sign out of both Azure AD and any Microsoft accounts you are currently signed in as in your current browser session.

The next step will redirect you to Microsoft SSO to sign in with one of our test users, but if an active session to

another Microsoft account is detected, it will use that instead. Importing the SAML attributes will then fail, as this account is likely not associated with our Azure AD tenant.

In the ZPA admin portal, go to **Administration > IdP Configuration** and expand the IdP configuration you want to import the attributes from. Under **Import SAML Attributes**, select the domain associated with the IdP you want to import the attributes from and click **Import**.

The screenshot shows the 'IdP Configuration' page in the ZPA Admin Portal. The left sidebar contains navigation links: Dashboard, Diagnostics, Live Logs, Administration (highlighted), Search, and Zscaler App. The main content area has tabs for 'IdP Configuration', 'SAML Attributes', and 'Settings'. A table lists IdP configurations, with 'Azure AD - Oblivion Industries' selected. Below the table, the 'Import SAML Attributes' section is highlighted with a red box, showing a dropdown menu with 'oblivion.industries' and an 'Import' button. Other sections include 'Single Sign-On URL', 'Authentication Domains', 'Service Provider SAML Metadata for User SSO', 'Service Provider Certificate', 'Service Provider Entity ID', 'IdP Certificate', and 'SCIM'.

Name	Status	IdP Entity ID	Single Sign-On	Actions
Azure AD - Oblivion Industries	✓	https://sts.windows.net/[redacted]/	User	[edit] [delete]

Import SAML Attributes
oblivion.industries [dropdown] [Import]

Single Sign-On URL
https://login.microsoftonline.com/[redacted]/saml2

Authentication Domains
oblivion.industries

Service Provider SAML Metadata for User SSO

Service Provider Metadata	Service Provider Certificate
Download Metadata	Download Certificate

Service Provider URL
https://samlsp.private.zscaler.com/auth/[redacted]/sso

Service Provider Entity ID
https://samlsp.private.zscaler.com/auth/metadata/[redacted]

IdP Certificate

Common Name	Serial Number
Microsoft Azure Federated SSO Certificate	[redacted]

Created On
Sunday, May 10 2020 6:20:45 pm

Expires On
Wednesday, May 10 2023 6:20:41 pm

SCIM

SCIM Sync
✓ Enabled

SCIM Service Provider Endpoint
https://scim1.private.zscaler.com/scim/1/[redacted]/v2

Sign in using the credentials of a user that is authorized to use ZPA (ie: a user associated with the ZPA enterprise application). This will test your SSO config so far:

- If you encounter a Microsoft error (and you're trying this in an incognito or private tab), this indicates an issue with the Microsoft side of the config. Check to see whether the user you are attempting to sign in as has been added to the enterprise application as an authorized user. Additionally, check that your SAML config is correct.
- If you encounter a Zscaler branded error, this indicates an issue with the config on the Zscaler side. Note down the error code in the bottom left and check it against a [list of error codes here](#).

If you have a successful login, the SAML JSON response will be captured and used to discover available SAML attributes to import:

Dashboard
Diagnostics
Live Logs
Administration
Search
Zscaler App

Import SAML Attributes

Name	SAML Attribute Name
Tenant ID	http://schemas.microsoft.com/identity/claims/tenantid
Object ID	http://schemas.microsoft.com/identity/claims/objectidentifier
Full Name	http://schemas.microsoft.com/identity/claims/displayname
http://schemas.microsoft.com/ws/2008/06/identity/claims/groups_Azure AD - Oblivion Industries	http://schemas.microsoft.com/ws/2008/06/identity/claims/groups
http://schemas.microsoft.com/identity/claims/identityprovider_Azure AD - Oblivion Industries	http://schemas.microsoft.com/identity/claims/identityprovider
http://schemas.microsoft.com/claims/authnmethodsreferences_Azure AD - Oblivion Industries	http://schemas.microsoft.com/claims/authnmethodsreferences
http://schemas.microsoft.com/identity/claims/agegroup_Azure AD - Oblivion Industries	http://schemas.microsoft.com/identity/claims/agegroup
First Name	http://schemas.xmlsoap.org/ws/2005/05/identity/claims/givenname
http://schemas.xmlsoap.org/ws/2005/05/identity/claims/surname_Azure AD - Oblivion Industries	http://schemas.xmlsoap.org/ws/2005/05/identity/claims/surname
http://schemas.xmlsoap.org/ws/2005/05/identity/claims/emailaddress_Azure AD - Oblivion Industries	http://schemas.xmlsoap.org/ws/2005/05/identity/claims/emailaddress
Username	http://schemas.xmlsoap.org/ws/2005/05/identity/claims/name

Save
Cancel

Import SAML JSON

```

{
  "nameid": "nathan@oblivion.industries",
  "orgid": "144133171103924224",
  "idpEntityID": "https://sts.windows.net/[redacted]",
  "idpid": "[redacted]5",
  "saml_attributes": {
    "http://schemas.microsoft.com/identity/claims/tenantid": "[redacted]",
    "http://schemas.microsoft.com/identity/claims/objectidentifier": "[redacted]",
    "http://schemas.microsoft.com/identity/claims/displayname": "Nathan Catania",
    "http://schemas.microsoft.com/ws/2008/06/identity/claims/groups": [

```

Anything in the left column, **Name**, you should overwrite with a descriptive name. **What you type here will appear when creating an access policy**, so Full Name is a lot easier to understand than ...claims_displayname_Azure AD is.

NB: If you have multiple IdPs configured, the name given to the attribute must be unique amongst all of them or it will not import.

Some important mappings you might want to be aware of:

SAML Attribute Name	What you should call it
displayname	Full Name
groups	Group or memberOf
givenname	First Name

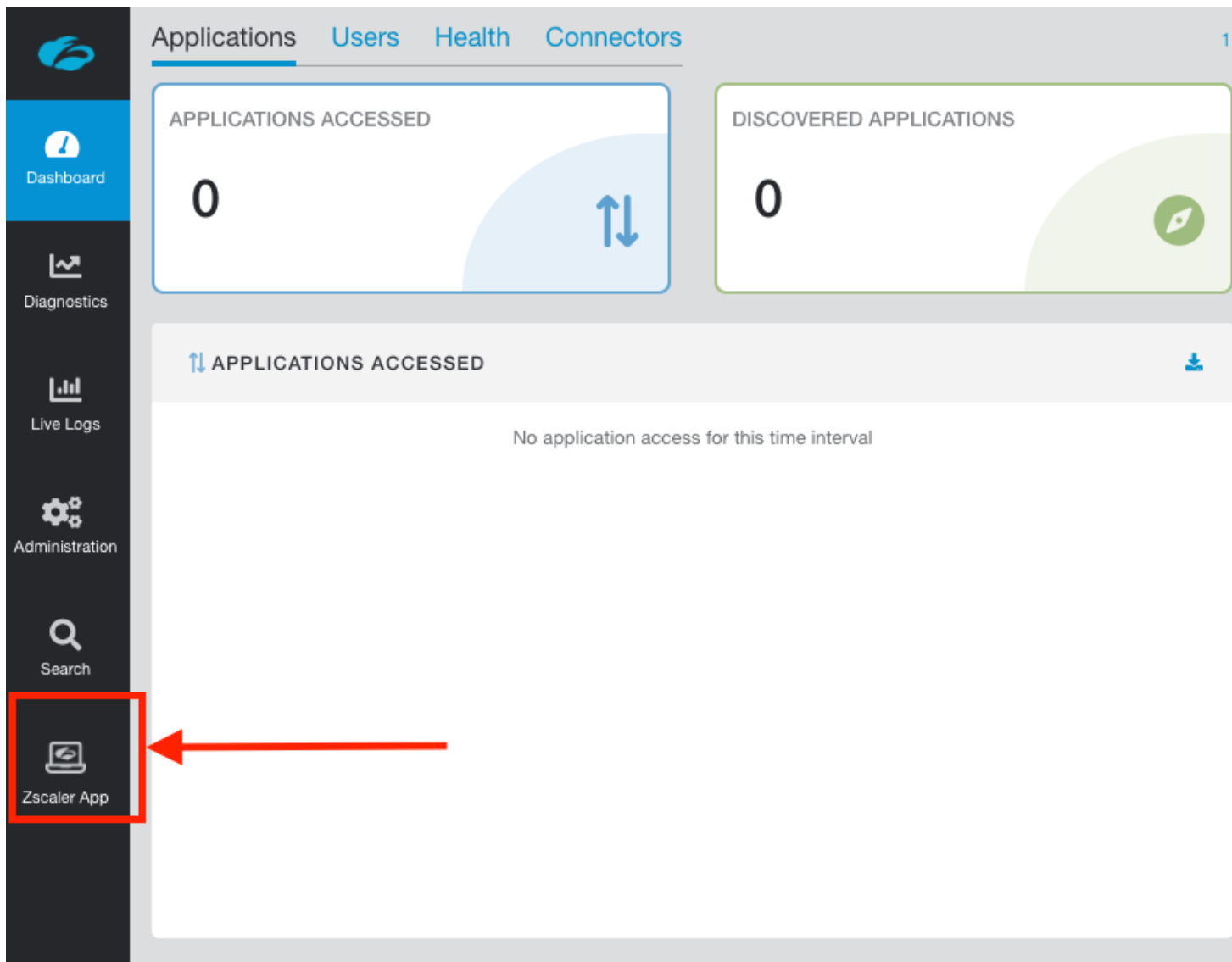
SAML Attribute Name	What you should call it
name	Username (eg: user@example.com)
surname	Surname
emailaddress	Email Address

You can also use the JSON displayed at the bottom of the page to work out what things should be called based on the captured data.

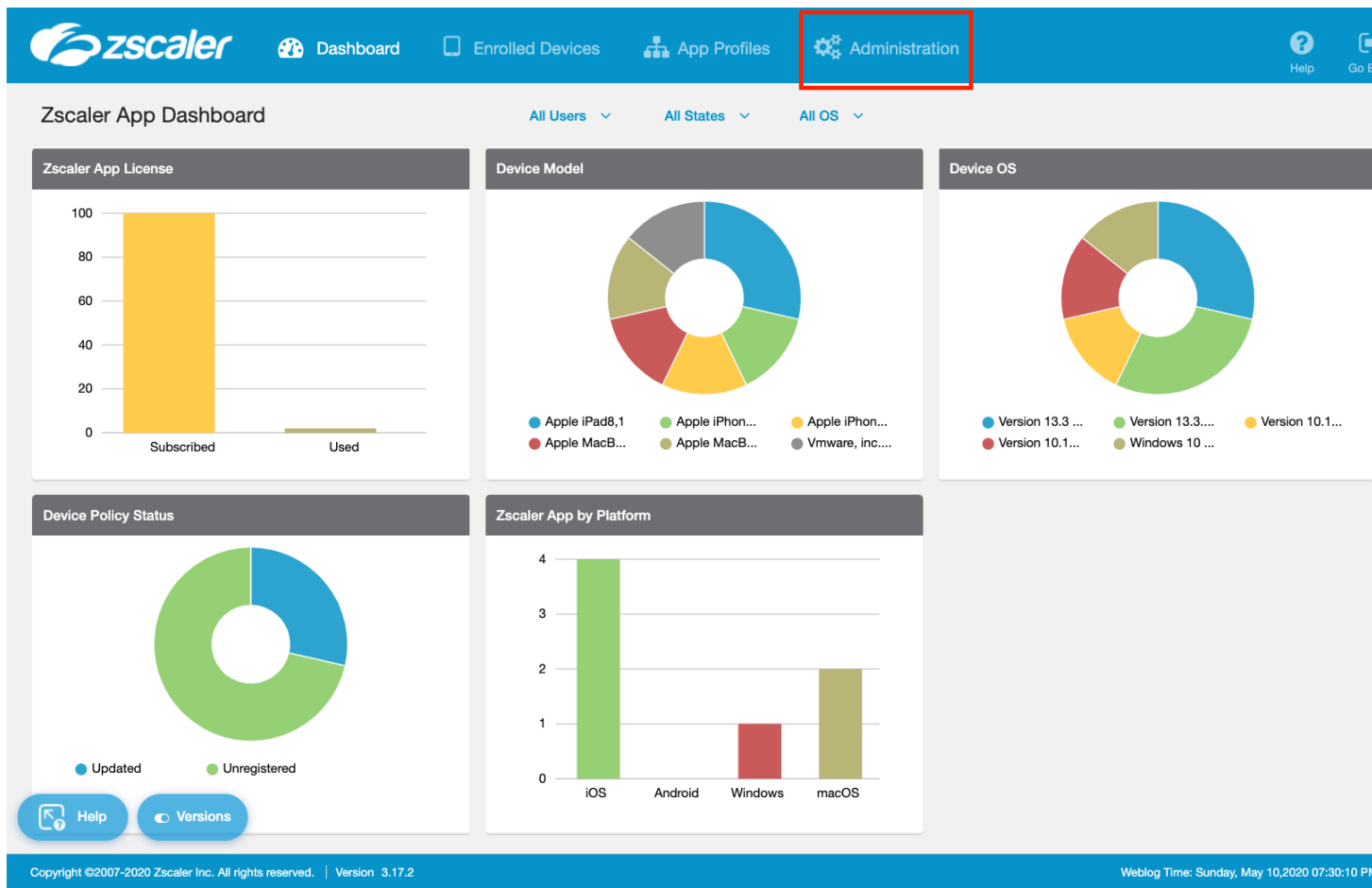
Note: Only SAML attributes that have not been imported are displayed in the table. If the table is blank or some fields are missing, check you haven't already imported the attribute you're after.

When you are done, click **Save** to finish the import.

You will now be able to create an Access Policy using any of the imported SAML attributes:

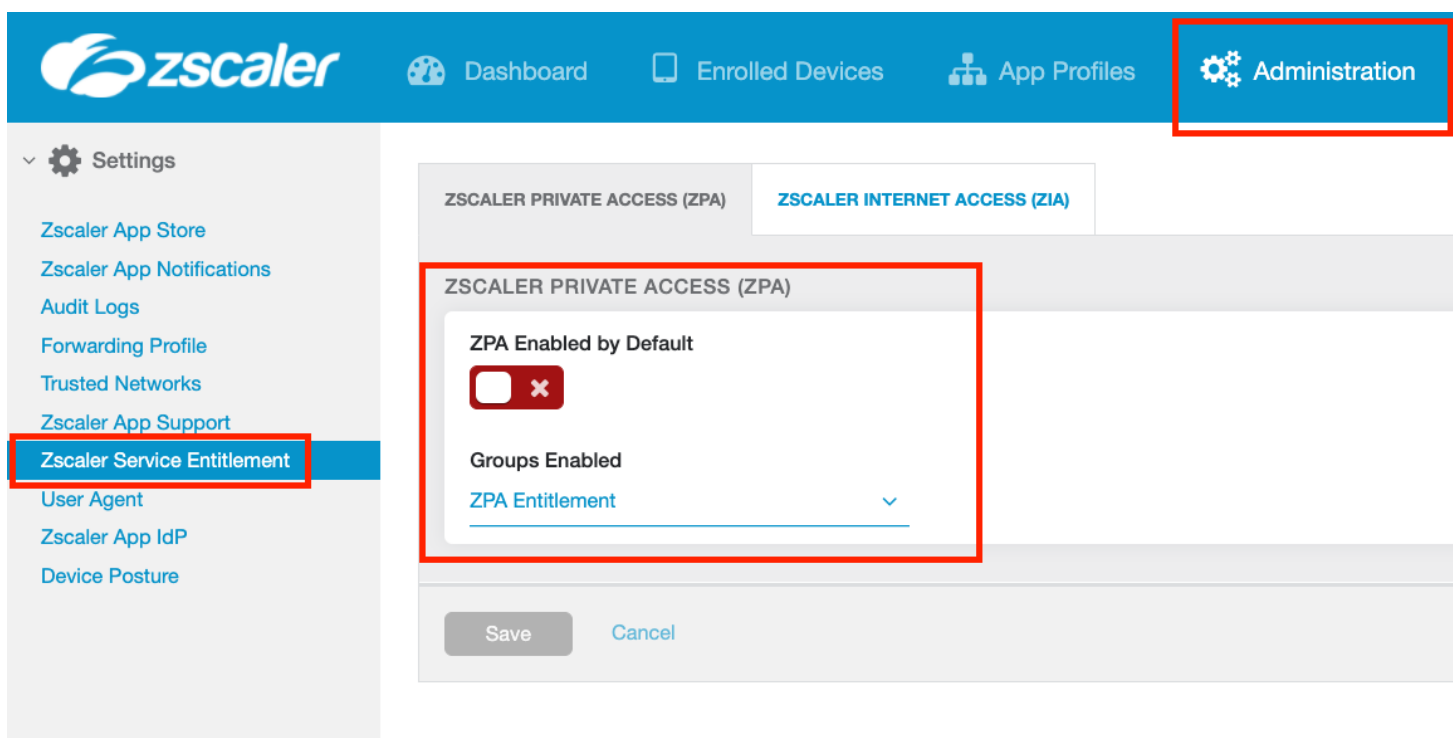


Once in the Zscaler App Portal go to **Administration > Zscaler Service Entitlement**.



IMPORTANT: If you have a brand new ZPA portal, you will not see the option for **Zscaler Service Entitlement** in the Zscaler App Portal. You need to [raise a ticket with support](#) to have this enabled, and can use the following script:

Hi team. IdP is now configured in ZPA. Please enable ZPA in the Zscaler App Portal at your earliest convenience. Thank you.



You can choose to enable ZPA in the Zscaler App globally for all users, or only allow access to specific groups.

IMPORTANT: Group membership is synced from ZIA. If you've changed group membership or created a new group in Azure AD, you must wait for it to sync to ZIA with SCIM AND wait for it to sync from ZIA to the Zscaler App portal (you can also initiate a manual pull from ZIA under Administration > Zscaler App Support > Advanced Configuration)

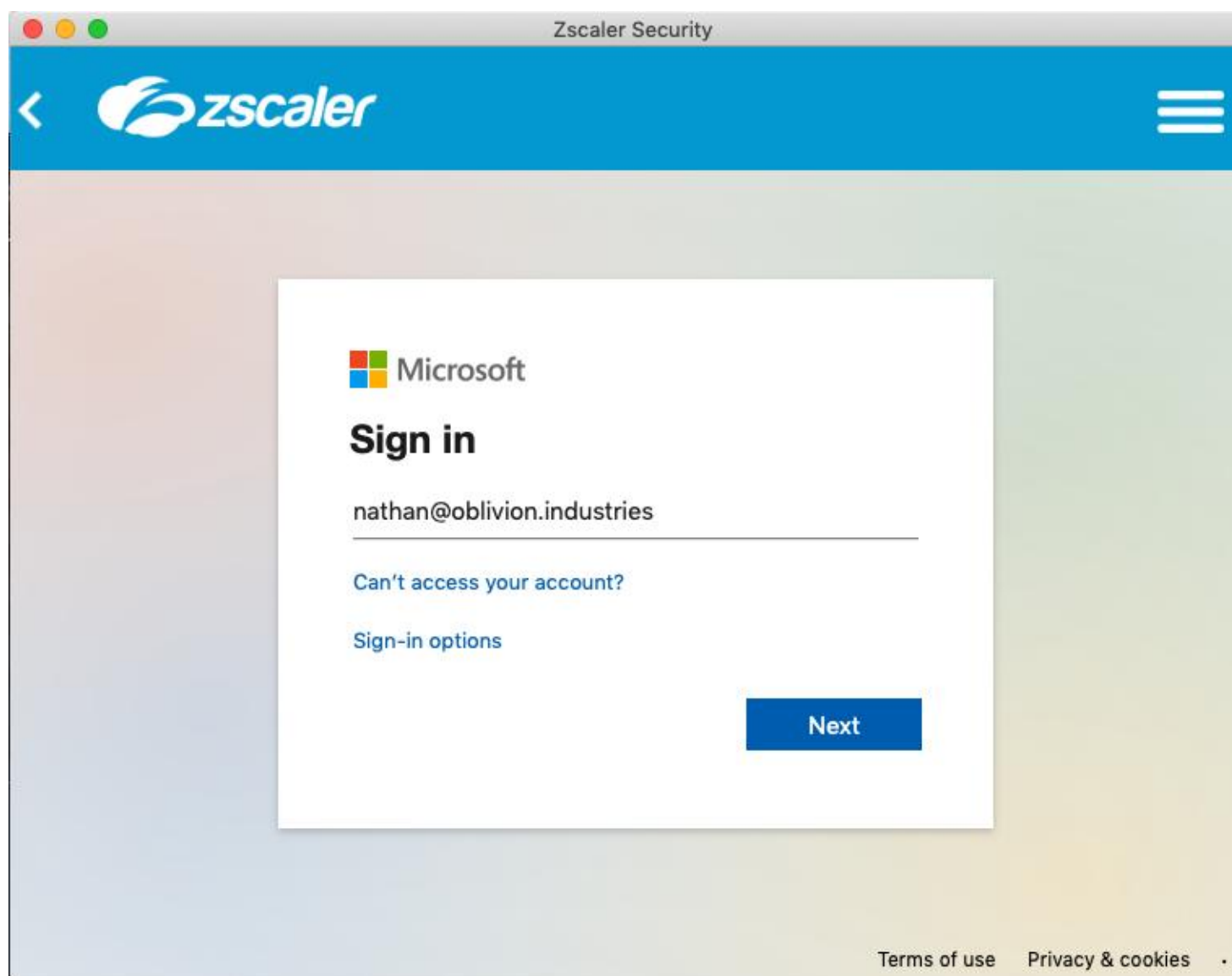
Select whether to enable ZPA for all users (check the **ZPA Enabled by Default** option), or select users by specifying a group.

Click **Save** when you are finished and return to the ZPA admin portal.

Test the Azure AD configuration

We're going to test everything that we've done so far. Open Z-App (sign-out if you're already signed in with another user) and sign in with the credentials of one of your Azure AD users that you assigned to the ZPA Enterprise Application.

When you enter the username, Z-App should now redirect you to sign-in with Microsoft. Enter valid user credentials and test the sign-in.



If everything is working well, and you've enabled ZPA in the Zscaler App (Zscaler Service Entitlement), the user will be signed in to both ZIA and ZPA (provided Azure AD has been used for the IdP for both). Z-App will minimize itself to your system tray or menu-bar. If you open it, you'll see your user is authenticated.

The screenshot shows the 'IdP Configuration' page in the Zscaler Admin Portal. The left sidebar contains navigation links: Dashboard, Diagnostics, Live Logs, Administration (highlighted), Search, and Zscaler App. The main content area has tabs for 'IdP Configuration', 'SAML Attributes', and 'Settings'. A table lists IdP configurations, with 'Azure AD - Oblivion Industries' selected. Below the table, the configuration details are shown, including the Single Sign-On URL, Authentication Domains, and SAML Attributes. A red box highlights the 'Import SAML Attributes' button. The 'Service Provider SAML Metadata for User SSO' section includes links to download metadata and certificates, and the 'IdP Certificate' section shows the certificate details. The 'SCIM' section shows that SCIM Sync is enabled and provides the SCIM Service Provider Endpoint.

Name	Status	IdP Entity ID	Single Sign-On	Actions
Azure AD - Oblivion Industries	✓	https://sts.windows.net/[redacted]/	User	[edit] [delete]

Single Sign-On URL
https://login.microsoftonline.com/[redacted]/saml2

Authentication Domains
oblivion.industries

Import SAML Attributes
oblivion.industries [dropdown] [Import]

Service Provider SAML Metadata for User SSO

Service Provider Metadata
[Download Metadata]

Service Provider URL
https://samlsp.private.zscaler.com/auth/[redacted]/sso

Service Provider Certificate
[Download Certificate]

Service Provider Entity ID
https://samlsp.private.zscaler.com/auth/metadata/[redacted]

IdP Certificate

Common Name
Microsoft Azure Federated SSO Certificate

Created On
Sunday, May 10 2020 6:20:45 pm

Serial Number
[redacted]

Expires On
Wednesday, May 10 2023 6:20:41 pm

SCIM

SCIM Sync
✓ Enabled

SCIM Service Provider Endpoint
https://scim1.private.zscaler.com/scim/1/[redacted]/v2

Testing SSO in a Browser

You can test the IdP configuration outside of the ZPA admin portal using the following URL. Replace EXAMPLE.COM with your company's domain name as per Azure AD.

IMPORTANT: Test the URL in a Private/Incognito window.

<https://samlsp.private.zscaler.com/auth/v2/login?ssotype=test&domain=EXAMPLE.COM>

Troubleshooting Sign-in

If the Microsoft SSO fails, check the config on the Azure AD side. This could be due to:

- The URLs in the SAML config are not correct.
- Your user has not been assigned to the enterprise application.

If sign in passes Microsoft SSO, but you get a Zscaler error (as per the image below), check the error code in the bottom right corner against the [list of error codes here](#). This indicates an issue with the config on the Zscaler side (typically something to do with the certificate you imported from Azure AD).

Name	SAML Attribute Name
Tenant ID	http://schemas.microsoft.com/identity/claims/tenantid
Object ID	http://schemas.microsoft.com/identity/claims/objectidentifier
Full Name	http://schemas.microsoft.com/identity/claims/displayname
http://schemas.microsoft.com/ws/2008/06/identity/claims/groups	http://schemas.microsoft.com/ws/2008/06/identity/claims/groups
http://schemas.microsoft.com/identity/claims/identityprovider	http://schemas.microsoft.com/identity/claims/identityprovider
http://schemas.microsoft.com/claims/authnmethodsreferences	http://schemas.microsoft.com/claims/authnmethodsreferences
http://schemas.microsoft.com/identity/claims/agegroup	http://schemas.microsoft.com/identity/claims/agegroup
First Name	http://schemas.xmlsoap.org/ws/2005/05/identity/claims/givenname
http://schemas.xmlsoap.org/ws/2005/05/identity/claims/surname	http://schemas.xmlsoap.org/ws/2005/05/identity/claims/surname
http://schemas.xmlsoap.org/ws/2005/05/identity/claims/emailaddress	http://schemas.xmlsoap.org/ws/2005/05/identity/claims/emailaddress
Username	http://schemas.xmlsoap.org/ws/2005/05/identity/claims/name

Save Cancel

Import SAML JSON

```
{
  "nameid": "nathan@oblivion.industries",
  "orgid": "144133171103924224",
  "idpEntityID": "https://sts.windows.net/[redacted]",
  "idpid": "[redacted]5",
  "saml_attributes": {
    "http://schemas.microsoft.com/identity/claims/tenantid": "[redacted]",
    "http://schemas.microsoft.com/identity/claims/objectidentifier": "[redacted]",
    "http://schemas.microsoft.com/identity/claims/displayname": "Nathan Catania",
    "http://schemas.microsoft.com/ws/2008/06/identity/claims/groups": [
```

Finish

If all went well, you should now have your ZPA tenant integrated with Azure AD with SAML-based SSO and SCIM user provisioning.

- From this image we can see the usernames and the passwords getting auto filled. This is a potential vulnerability as this can be a doorway for attackers.

Responsibilities:

- Implement secure password storage mechanisms, such as using strong and salted cryptographic hashing algorithms like bcrypt or scrypt.
- Implement robust encryption key management practices, including secure key storage, key rotation, and access controls.
- Utilize well-established and proven cryptographic libraries and algorithms for encryption and decryption.
- Use secure encryption protocols, like TLS/SSL, for data transmission over networks to protect data in transit.

2. Vulnerability name: Cross-site scripting (DOM based XSS)

CWE: 79

OWASP category: A03: Injections

Description: It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser.

DOM-XSS: DOM stands for Document Object Model and is a programming interface for HTML and XML documents. It represents that programs can change the document structure, style and content. A webpage is a document and this document can be either displayed in the browser window or as the HTML Source.

Business Impact: The application stores dangerous data in a database, message forum, visitor log, or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area displayed to many users or particularly interesting users. Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data belonging to the user.

- We will try to perform the DOM XSS attack which is displaying the source code of the webpage.
- Now we can access the source code of the webpage in two ways by clicking on the inspect option or by simply pressing Ctrl + U.

```

Line wrap
1 <!DOCTYPE html>
2
3 <html lang="en">
4 <head>
5 <meta charset="utf-8">
6 <meta http-equiv="X-UA-Compatible" content="IE=edge,chrome=1">
7 <meta name="viewport" content="width=device-width, initial-scale=1.0">
8 <meta http-equiv="Content-Type" content="text/html; charset=utf-8" />
9 <link rel="alternate" type="application/rss+xml" title="VIT RSS" href="https://vit.ac.in/rss.xml" />
10 <link rel="shortcut icon" href="https://vit.ac.in/sites/all/themes/vittheme/favicon.ico" type="image/vnd.microsoft.icon" />
11 <meta name="description" content="One of the Best Educational Institution in India. Maintained an excellent placement records and gives training students to get
12 <meta name="keywords" content="vit, vellore institute of technology, viteee, vit university, vit vellore, top engineering colleges in india, top 10 engineering c
13 <meta name="generator" content="Drupal" />
14 <link rel="canonical" href="https://vit.ac.in/" />
15 <meta property="og:site_name" content="Vellore Institute of Technology" />
16 <meta property="og:type" content="website" />
17 <meta property="og:url" content="https://vit.ac.in/" />
18 <meta property="og:title" content="Vellore Institute of Technology | A Place to Learn, Chance to grow" />
19 <meta property="og:description" content="One of the Best Educational Institution in India. Maintained an excellent placement records and gives training students
20 <meta property="og:image" content="https://vit.ac.in/sites/all/themes/vittheme/images/og-vit-updated.jpg" />
21 <meta property="og:image:url" content="https://vit.ac.in/sites/all/themes/vittheme/images/og-vit-updated.jpg" />
22 <meta name="dcterms:title" content="VIT" />
23 <meta name="dcterms:type" content="Text" />
24 <meta name="dcterms:format" content="text/html" />
25 <meta name="dcterms:identifier" content="https://vit.ac.in/" />
26 <meta name="facebook-domain-verification" content="9bbekg8iy8yy964rj4euqpwennaxoc" />
27 <title>Vellore Institute of Technology | A Place to Learn, Chance to grow</title>
28 <style type="text/css" media="screen">
29 @import url("https://vit.ac.in/sites/all/themes/vittheme/css/bootstrap.min.css?rdiap4");
30 @import url("https://vit.ac.in/sites/all/themes/vittheme/css/style.css?rdiap4");
31 @import url("https://vit.ac.in/sites/all/themes/vittheme/css/theme.css?rdiap4");
32 @import url("https://vit.ac.in/sites/all/themes/vittheme/css/combine.css?rdiap4");
33 @import url("https://vit.ac.in/sites/all/themes/vittheme/css/responsive.css?rdiap4");
34 </style>
35 <meta name="author" content="VIT University">
36 <meta http-equiv="Content-Type" content="text/html; charset=utf-8">
37 <meta name="language" content="English">
38 <meta name="copyright" content="https://vit.ac.in/" />
39 <meta name="geo.region" content="IN-TN" />
40 <meta name="geo.placename" content="Vellore" />
41 <meta name="geo.position" content="12.068812, 79.155017" />

```

- Many programs can help to alter this source code to find vulnerable spots.
- So, getting access to the source code of the webpage can lead to potential vulnerabilities as attackers can abuse the loop holes, exploits, vulnerabilities in the code and act accordingly.

Recommendation:

- When interacting with databases, use parameterized statements or prepared statements to avoid SQL injection, which can be a vector for XSS.
- Sanitize and validate all user inputs and ensure that any data displayed on the web page is properly encoded. Use output encoding libraries or functions to prevent script injection.
- Avoid using DOM manipulation methods that can introduce vulnerabilities. Be cautious with inner HTML, document. Write, and other methods that can execute scripts.
- Consider using well-established JavaScript frameworks and libraries that include built-in security features to prevent DOM XSS.

3. Vulnerability name: JQuery 1.2 < 3.5.0 Multiple XSS

CWE: 79

OWASP category: A03: Injections

Description: It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser.

Stored XSS (Persistent XSS): In this type of attack, the malicious script is permanently stored on a server or in a database, often in user-generated content like comments or forum posts. When other users view the affected

content, the script executes in their browsers, potentially compromising their data or sessions.

Reflected XSS: Reflected XSS occurs when the malicious script is embedded in a URL and immediately executed when a victim clicks on a manipulated link. The script is not stored on a server but is reflected off a web application, making it a one-time attack.

DOM-based XSS: DOM stands for Document Object Model and is a programming interface for HTML and XML documents. It represents that programs can change the document structure, style and content. A webpage is a document and this document can be either displayed in the browser window or as the HTML Source.

- We can access the source code of the jquery script that has been used in making the website, which can lead to multiple potential threats

Recommendation:

- The most important recommendation is to update to the latest version of jQuery. Newer versions typically include security patches, bug fixes, and performance improvements.
- Stay vigilant about jQuery updates and security announcements. Subscribe to relevant security mailing lists or forums to receive notifications about new releases.
- A Content Security Policy (CSP) can help mitigate various types of client-side vulnerabilities, including Cross-Site Scripting (XSS). Consider implementing a CSP in your web application to restrict the sources from which scripts can be loaded.
- If updating jQuery is not immediately possible, review your code for potential security issues. Ensure that you are escaping or sanitizing user inputs and that you are using safe coding practices to prevent XSS and other security vulnerabilities.

4. Vulnerability name: Information Exposure (Web Server HTTP Header Internal IP Disclosure)

CWE: 200

Description: The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information.

Business Impact: To effectively reduce the business consequences associated with CWE-200, organizations must prioritize the implementation of robust data access controls, secure data handling practices, and privacy protection measures. Regular security assessments and testing are instrumental in detecting and rectifying vulnerabilities related to information exposure. Furthermore, strict adherence to data protection regulations and industry standards is indispensable. These actions are of paramount importance in the protection of sensitive data, the preservation of user trust, and the upholding of the organization's esteemed reputation.

We will be using a tool to perform a scan on this website. (Nessus Scanner)

LOW

Web Server HTTP Header Internal IP Disclosure

Description

This may expose internal IP addresses that are usually hidden or masked behind a Network Address Translation (NAT) Firewall or proxy server.

There is a known issue with Microsoft IIS 4.0 doing this in its default configuration. This may also affect other web servers, web applications, web proxies, load balancers and through a variety of misconfigurations related to redirection.

Solution

Apply configuration suggested by vendor.

- Through this tool we can see the vulnerability in the scan along with detailed information regarding that vulnerability.

Nessus was able to exploit the issue using the following request :

```
GET / HTTP/1.0
Accept-Charset: iso-8859-1,utf-8;q=0.9,*;q=0.1
Accept-Language: en
Connection: Close
User-Agent: Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 5.1; Trident/4.0)
Pragma: no-cache
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, image/png, */*
```

This produced the following truncated output (limited to 10 lines) :

```
----- snip -----
Location: https://10.10.7.35/
Content-Length: 0

----- snip -----
less...
```

- Now this image tells us the internal IP address of the target website, the attackers can easily exploit the website using the ip address.

Recommendation:

- Carefully review your web server's configuration files (e.g., Apache's httpd.conf, Nginx's nginx.conf) to identify anywhere internal IP addresses are inadvertently included in HTTP response headers.
- Remove or replace any internal IP addresses in the server configuration files with appropriate placeholders or public IP addresses.
- If you use proxy servers or load balancers, ensure that they are configured to hide internal IP addresses and only expose public IP addresses instances in HTTP headers.
- Review and configure HTTP response headers (e.g., Server, X-Powered-By) to limit the exposure of server and technology details, reducing the potential for disclosing internal IP addresses.

5. Vulnerability name: Information Exposure (SSL (Multiple Issues))

CWE: 200

Business Impact: To effectively reduce the business consequences arising from SSL issues, organizations must give utmost priority to establishing a secure SSL/TLS configuration, conducting frequent security assessments and testing. Adhering to data protection regulations and industry standards is imperative. These actions play a pivotal role in safeguarding sensitive data, upholding user trust, and safeguarding the organization's esteemed reputation.

- We will be using a tool to perform a scan on this website. (Nessus Scanner)

Vulnerabilities 22				
Search Vulnerabilities  4 Vulnerabilities				
☐ Sev ▾	CVSS ▾	VPR ▾	Name ▴	Family ▴
☐ INFO			SSL Certificate Information	General
☐ INFO			SSL Cipher Block Chaining Cipher Suites Sup...	General
☐ INFO			SSL Cipher Suites Supported	General
☐ INFO			SSL Perfect Forward Secrecy Cipher Suites S...	General

- Through this tool we can see the vulnerability in the scan along with detailed information regarding that vulnerability.

INFO SSL Certificate Information

Description

This plugin connects to every SSL-related port and attempts to extract and dump the X.509 certificate.

Output

```
Subject Name:

Common Name: *.vit.ac.in

Issuer Name:

Country: GB
State/Province: Greater Manchester
Locality: Salford
Organization: Sectigo Limited
Common Name: Sectigo RSA Domain Validation Secure Server CA

Serial Number: 46 53 67 B6 23 C5 BE EE B9 6E E2 C0 5F 46 C9 F7

Version: 3

Signature Algorithm: SHA-256 With RSA Encryption

Not Valid Before: Sep 04 00:00:00 2023 GMT
Not Valid After: Aug 03 23:59:59 2024 GMT

Public Key Info:

Algorithm: RSA Encryption
Key Length: 2048 bits
```

- Now this image tells us the type of encryption used in the target website with the public key details as well.

INFO SSL Cipher Block Chaining Cipher Suites Supported

Description

The remote host supports the use of SSL ciphers that operate in Cipher Block Chaining (CBC) mode. These cipher suites offer additional security over Electronic Codebook (ECB) mode, but have the potential to leak information if used improperly.

INFO SSL Cipher Suites Supported

Description

This plugin detects which SSL ciphers are supported by the remote service for encrypting communications.

INFO SSL Perfect Forward Secrecy Cipher Suites Supported

Description

The remote host supports the use of SSL ciphers that offer Perfect Forward Secrecy (PFS) encryption. These cipher suites ensure that recorded SSL traffic cannot be broken at a future date if the server's private key is compromised.

- Through this we can understand what different types of encryptions are used in the website.

Here is the list of SSL CBC ciphers supported by the remote server :

High Strength Ciphers (>= 112-bit key)

Name	Code	KEX	Auth	Encryption	MAC
-----	-----	---	----	-----	---
ECDHE-RSA-CAMELLIA-CBC-128	0xC0, 0x76	ECDH	RSA	Camellia-CBC (128)	SHA256
ECDHE-RSA-CAMELLIA-CBC-256	0xC0, 0x77	ECDH	RSA	Camellia-CBC (256)	SHA384
DHE-RSA-AES128-SHA	0x00, 0x33	DH	RSA	AES-CBC (128)	SHA1
DHE-RSA-AES256-SHA	0x00, 0x39	DH	RSA	AES-CBC (256)	SHA1
DHE-RSA-CAMELLIA128-SHA	0x00, 0x45	DH	RSA	Camellia-CBC (128)	SHA1
DHE-RSA-CAMELLIA256-SHA	0x00, 0x88	DH	RSA	Camellia-CBC (256)	SHA1
DHE-RSA-SEED-SHA	0x00, 0x9A	DH	RSA	SEED-CBC (128)	SHA1
ECDHE-RSA-AES128-SHA	0xC0, 0x13	ECDH	RSA	AES-CBC (128)	SHA1
ECDHE-RSA-AES256-SHA	0xC0, 0x14	ECDH	RSA	AES-CBC (256)	SHA1
AES128-SHA	0x00, 0x2F	RSA	RSA	AES-CBC (128)	SHA1
AES256-SHA	0x00, 0x35	RSA	RSA	AES-CBC (256)	SHA1
CAMELLIA128-SHA	0x00, 0x41	RSA	RSA	Camellia-CBC (128)	SHA1
CAMELLIA256-SHA	0x00, 0x84	RSA	RSA	Camellia-CBC (256)	SHA1
SEED-SHA	0x00, 0x96	RSA	RSA	SEED-CBC (128)	SHA1
DHE-RSA-AES128-SHA256	0x00, 0x67	DH	RSA	AES-CBC (128)	SHA256
DHE-RSA-AES256-SHA256	0x00, 0x6B	DH	RSA	AES-CBC (256)	SHA256
DHE-RSA-CAMELLIA128-SHA256	0x00, 0xBE	DH	RSA	Camellia-CBC (128)	SHA256
DHE-RSA-CAMELLIA256-SHA256	0x00, 0xC4	DH	RSA	Camellia-CBC (256)	SHA256
ECDHE-RSA-AES128-SHA256	0xC0, 0x27	ECDH	RSA	AES-CBC (128)	SHA256
ECDHE-RSA-AES256-SHA384	0xC0, 0x28	ECDH	RSA	AES-CBC (256)	SHA384
RSA-AES128-SHA256	0x00, 0x3C	RSA	RSA	AES-CBC (128)	SHA256
RSA-AES256-SHA256	0x00, 0x3D	RSA	RSA	AES-CBC (256)	SHA256
RSA-CAMELLIA128-SHA256	0x00, 0xBA	RSA	RSA	Camellia-CBC (128)	SHA256
RSA-CAMELLIA256-SHA256	0x00, 0xC0	RSA	RSA	Camellia-CBC (256)	SHA256

Recommendation:

- Configure your web server to use the latest and strongest encryption protocols, such as TLS 1.2 or higher. Disable outdated and insecure protocols like SSLv3.
- Implement Perfect Forward Secrecy to ensure that even if the private key is compromised, previously recorded encrypted traffic remains secure.
- Safeguard your private keys by storing them securely and ensuring they are not accessible to unauthorized parties.

6. Vulnerability name: Information Exposure (HTTP (Multiple Issues))

CWE: 200

Description: The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information.

Business Impact: To effectively reduce the business repercussions of HTTP issues, organizations must place a strong emphasis on ongoing monitoring and enhancement of their web services. Consistent performance testing and security assessments, coupled with proactive maintenance, are indispensable to ensure a seamless and secure online presence. Furthermore, offering timely and helpful customer support and transparent communication when HTTP issues arise is instrumental in mitigating adverse effects on user trust and satisfaction.

Steps to Reproduce:

- We will be using a tool to perform a scan on this website. (Nessus Scanner)

Search Vulnerabilities

2 Vulnerabilities

Sev

CVSS

VPR

Name

Family

INFO

HyperText Transfer Protocol (HTTP) Informa...

Web Servers

INFO

HTTP Server Type and Version

Web Servers

INFO HyperText Transfer Protocol (HTTP) Information

Description

This test gives some information about the remote HTTP protocol - the version used, whether HTTP Keep-Alive and HTTP pipelining are enabled, etc...

This test is informational only and does not denote any security problem.

```
Response Code : HTTP/1.1 307 Moved Temporarily

Protocol version : HTTP/1.1|
SSL : no
Keep-Alive : no
Options allowed : (Not implemented)
Headers :

    Location: https://vit.ac.in/
    Content-Length: 0

Response Body :
```

- From the above image we can see the details of protocol version of the website and its status.

INFO HTTP Server Type and Version

Description

This plugin attempts to determine the type and the version of the remote web server.

The remote web server type is :
Apache

- We can see the type of remote webserver here.
- All this information can lead to multiple potential threats such as data breaches by hackers.

Recommendation:

- Minimize the use of unnecessary redirects as they can increase page load times. Use 301 (permanent) redirects for consistent content.
- Regularly check for broken links (404 errors) and fix them to ensure users find the content they are looking for.
- Properly configure CORS to control which domains are allowed to make requests to your server.
- Migrate your website from HTTP to HTTPS to ensure secure data transmission. This is essential for user trust and search engine ranking.

7. Vulnerability name: OS Identification Failed

CWE: 200

Description: OS identification failed refers to the inability of a system or software to accurately recognize the operating system running on a device or network. This issue can result from various factors such as misconfigurations, network issues, or compatibility problems. The failure to identify the OS correctly can lead to security vulnerabilities, hindered software updates, and potential operational disruptions. Resolving this issue promptly is crucial to ensure accurate security measures and the smooth functioning of the system or network.

Business Impact: It may lead to operational inefficiencies, potential security vulnerabilities, non-compliance with industry regulations, and a possible loss of customer trust. Such failures could result in downtime, compromised data security, regulatory penalties, and damage to the company's reputation, potentially leading to financial losses and decreased market competitiveness. Resolving this issue promptly is crucial to mitigate these risks and maintain the smooth functioning of the organization.

- We will be using a tool to perform a scan on this website. (Nessus Scanner)

Description

Using a combination of remote probes (TCP/IP, SMB, HTTP, NTP, SNMP, etc), it was possible to gather one or more fingerprints from the remote system. Unfortunately, though, Nessus does not currently know how to use them to identify the overall system.

```
If you think these signatures would help us improve OS fingerprinting,
please send them to :
```

```
os-signatures@nessus.org
```

```
Be sure to include a brief description of the device itself, such as
the actual operating system or product / model names.
```

```
SSLcert:::i/CN:Sectigo RSA Domain Validation Secure Server CAi/O:Sectigo
Limiteds/CN:*.vit.ac.in
02ed800c9e7f764615d7e9ffd74b643a6cace446
```

```
SinFP:::
```

```
P1:B10113:F0x12:W65535:00204ffff:M496:
```

```
P2:B10113:F0x12:W65535:00204ffff01030303010104020101080affffffff44454144:M1300:
```

```
P3:B00000:F0x00:W0:00:M0
```

```
P4:190701_7_p=80R
```

- From the above image, we can see the details of SSL Certificate Information and SinFP Information.
- If it is released without proper authorization, it can lead to serious consequences such as increased vulnerability to cyberattacks, potential data breaches, privacy violations, and legal repercussions.
- All this information can lead to multiple potential threats such as data breaches by hackers.

Recommendations:

- Employ robust data encryption methods for both data in transit and at rest.
- Implement strict access controls and authentication protocols to limit unauthorized access.
- Use secure communication channels, such as encrypted email and file transfer protocols.
- Provide comprehensive training to employees on data security best practices.
- Conduct regular security audits and assessments to identify and address vulnerabilities promptly.
- Ensure compliance with relevant data protection regulations and have a well-defined incident response plan in place.

NESSUS SCAN REPORT:



Vulnerabilities

Total: 28

SEVERITY	CVSS V3.0	VPR SCORE	PLUGIN	NAME
MEDIUM	6.5	-	142960	HSTS Missing From HTTPS Server (RFC 6797)
MEDIUM	6.5	-	104743	TLS Version 1.0 Protocol Detection
MEDIUM	6.5	-	157288	TLS Version 1.1 Protocol Deprecated
LOW	3.7	4.5	83875	SSL/TLS Diffie-Hellman Modulus <= 1024 Bits (Logjam)
INFO	N/A	-	46180	Additional DNS Hostnames
INFO	N/A	-	39446	Apache Tomcat Detection
INFO	N/A	-	45590	Common Platform Enumeration (CPE)
INFO	N/A	-	54615	Device Type
INFO	N/A	-	84502	HSTS Missing From HTTPS Server
INFO	N/A	-	10107	HTTP Server Type and Version
INFO	N/A	-	24260	HyperText Transfer Protocol (HTTP) Information
INFO	N/A	-	11219	Nessus SYN scanner
INFO	N/A	-	19506	Nessus Scan Information
INFO	N/A	-	11936	OS Identification
INFO	N/A	-	10919	Open Port Re-check
INFO	N/A	-	56984	SSL / TLS Versions Supported
INFO	N/A	-	10863	SSL Certificate Information
INFO	N/A	-	95631	SSL Certificate Signed Using Weak Hashing Algorithm (Known CA)
INFO	N/A	-	70544	SSL Cipher Block Chaining Cipher Suites Supported

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INFO	N/A	-	21643	SSL Cipher Suites Supported
INFO	N/A	-	57041	SSL Perfect Forward Secrecy Cipher Suites Supported
INFO	N/A	-	94761	SSL Root Certification Authority Certificate Information
INFO	N/A	-	156899	SSL/TLS Recommended Cipher Suites
INFO	N/A	-	22964	Service Detection
INFO	N/A	-	25220	TCP/IP Timestamps Supported
INFO	N/A	-	121010	TLS Version 1.1 Protocol Detection
INFO	N/A	-	136318	TLS Version 1.2 Protocol Detection
INFO	N/A	-	10287	Traceroute Information

* indicates the v3.0 score was not available; the v2.0 score is shown



Vulnerabilities

Total: 28

SEVERITY	CVSS V3.0	VPR SCORE	PLUGIN	NAME
MEDIUM	6.1	5.7	136929	JQuery 1.2 < 3.5.0 Multiple XSS
LOW	3.1	2.2	10759	Web Server HTTP Header Internal IP Disclosure
INFO	N/A	-	48204	Apache HTTP Server Version
INFO	N/A	-	45590	Common Platform Enumeration (CPE)
INFO	N/A	-	10107	HTTP Server Type and Version
INFO	N/A	-	24260	HyperText Transfer Protocol (HTTP) Information
INFO	N/A	-	106658	JQuery Detection
INFO	N/A	-	11219	Nessus SYN scanner
INFO	N/A	-	19506	Nessus Scan Information
INFO	N/A	-	42823	Non-compliant Strict Transport Security (STS)
INFO	N/A	-	50350	OS Identification Failed
INFO	N/A	-	66334	Patch Report
INFO	N/A	-	56984	SSL / TLS Versions Supported
INFO	N/A	-	10863	SSL Certificate Information
INFO	N/A	-	70544	SSL Cipher Block Chaining Cipher Suites Supported
INFO	N/A	-	21643	SSL Cipher Suites Supported
INFO	N/A	-	57041	SSL Perfect Forward Secrecy Cipher Suites Supported
INFO	N/A	-	94761	SSL Root Certification Authority Certificate Information
INFO	N/A	-	156899	SSL/TLS Recommended Cipher Suites

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INFO	N/A	-	22964	Service Detection
INFO	N/A	-	42822	Strict Transport Security (STS) Detection
INFO	N/A	-	25220	TCP/IP Timestamps Supported
INFO	N/A	-	84821	TLS ALPN Supported Protocol Enumeration
INFO	N/A	-	136318	TLS Version 1.2 Protocol Detection
INFO	N/A	-	10287	Traceroute Information
INFO	N/A	-	100669	Web Application Cookies Are Expired
INFO	N/A	-	10386	Web Server No 404 Error Code Check
INFO	N/A	-	10302	Web Server robots.txt Information Disclosure

* indicates the v3.0 score was not available; the v2.0 score is shown

"With the Nessus scan report now at our disposal, we're equipped with critical data on the existing vulnerabilities and potential threats. This invaluable information will serve as a foundation for our project planning phase, as we chart the course to develop our AI-driven incident response platform with a well-informed strategy and timeline."

Stage – 2

Overview: -

An overview of a Nessus scan typically involves the following key steps and aspects:

1. Preparation: Define Objectives: Determine the scope and objectives of the scan. What systems or networks do you want to assess, and what specific vulnerabilities are you looking for?

System Access: Ensure that you have the necessary permissions and access rights to scan the target systems or networks.

2. Configuration: Set Up Scan Policy: Create a scan policy that defines the parameters of the scan, such as the types of vulnerabilities to check for, scanning frequency, and specific configurations.

Define Target Assets: Specify the IP addresses, hostnames, or network ranges of the systems you want to scan.

3. Scanning: Initiate Scan: Start the Nessus scan using the configured policy and target assets. Nessus will systematically scan the target systems for vulnerabilities based on its plugins and policies.

4. Vulnerability Identification: Nessus performs a comprehensive assessment, checking for vulnerabilities, misconfigurations, open ports, and other security issues. As the scan progresses, it categorizes vulnerabilities by severity, making it easier to prioritize remediation efforts.

5. Reporting: Nessus generates detailed reports that list the vulnerabilities discovered during the scan. Reports typically include information about the severity of each vulnerability, recommended actions for remediation, and supporting evidence. These reports can be customized and exported in various formats, such as PDF, HTML, or CSV.

6. Remediation: Review Findings: IT and security teams review the Nessus scan reports to prioritize and address the identified vulnerabilities.

Remediation Steps: Take corrective actions to fix or mitigate the vulnerabilities. This may involve applying patches, changing configurations, or implementing security measures.

7. Follow-up: Regular Scanning: Conduct periodic Nessus scans to continuously monitor the security posture of the systems and networks. This helps identify new vulnerabilities and track progress in mitigating existing ones.

Integration: Integrate Nessus with other security tools or workflows to automate the vulnerability management process and streamline the response to new findings.

8. Compliance Auditing (if applicable): If compliance checks are part of your scan policy, review compliance results

and take actions to ensure systems meet the required standards and regulations.

their systems and to proactively identify potential vulnerabilities that could be exploited by attackers. Here's what I understand about Nessus scans:

1. **Vulnerability Assessment:** Nessus performs a vulnerability assessment by scanning target systems or networks to identify security issues, such as missing patches, misconfigurations, weak passwords, and other potential weaknesses that could be exploited by attackers.
2. **Network Scanning:** Nessus can scan a wide range of network devices, including servers, routers, switches, firewalls, and more. It examines network services, open ports, and software running on target systems.
3. **Plugin Architecture:** Nessus uses a plugin-based architecture, allowing it to check for a wide variety of vulnerabilities. These plugins are regularly updated to include the latest security checks and support for new vulnerabilities.
4. **Compliance Auditing:** It can also be used to assess systems against various compliance standards and regulatory requirements, helping organizations ensure that their systems meet specific security standards, like PCI DSS or CIS benchmarks.
5. **Customization:** Nessus scans can be customized to suit the specific needs of an organization. Users can define scan policies, select the target systems, and specify the types of vulnerabilities to scan for.
6. **Reporting:** Nessus provides detailed reports that highlight vulnerabilities, their severity, and remediation recommendations. These reports are essential for IT and security teams to prioritize and address vulnerabilities.
7. **Scanning Frequency:** Organizations often perform regular Nessus scans to continuously monitor and improve their security posture. This is crucial for staying ahead of evolving threats and vulnerabilities.
8. **Integration:** Nessus can be integrated with other security tools and platforms, allowing organizations to automate the scanning process and streamline their vulnerability management.
9. **Licensing:** Nessus offers both free and commercial versions. The free version is limited in its functionality, while the commercial version provides more features and support.
10. **User-Friendly Interface:** Nessus offers an easy-to-use web-based interface, making it accessible to a wide range of users, from security experts to less experienced IT professionals.

It's important to note that Nessus is a valuable tool for identifying vulnerabilities, but it doesn't fix these vulnerabilities itself. It's up to the organization's IT and security teams to take the necessary steps to remediate the identified issues to enhance their security.

S. No	Vulnerability Name	Severity	Plugins
1.	Cross-site scripting (stored)	Low	Web application Scanner
2.	Broken Access Control	Medium	SSH weak HMAC Algorithms Supported
3.	SQL Injection	Medium	N/A
4.	Broken Authentication	Low	HSTS Missing from Server
5.	Insecure Direct Object Reference	Info	Service Detection
6.	Security Misconfiguration	Low	SSL cipher Block Chaining cipher suites
7.	Cross-Site Request Forgery	Low	HTTP Form- Based Authentication Detected
8.	Cleartext Transmission of Sensitive Information	Medium	Cleartext Transmission of Sensitive Information over HTTP
9.	ClickJacking	Info	HTML Rendering Layer Vulnerability

10.	webserver allows password auto-completion	Low	SSL/TLS Diffie-Hellman Modulus<=1024 bits(Logjam)
-----	---	-----	---

Target Website: testfire.net Target

IP Address: 54.83.41.36 Report:

1. VULNERABILITY NAME: Cross site scripting

(stored) SEVERITY: Low

PLUGIN: web application scanner

PORT: N/A

DESCRIPTION: It occurs when a malicious script is injected directly into a vulnerable web application.

Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser.

SOLUTION:

- When interacting with databases, use parameterized statements or prepared statements to avoid SQL injection, which can be a vector for XSS.
- Sanitize and validate all user inputs and ensure that any data displayed on the web page properly encoded. Use output encoding libraries or functions to prevent script injection.

BUSINESS IMPACT: The application stores dangerous data in a database, message forum, visitor log,

or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area that is displayed to either many users or particularly interesting users. Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data belonging to the user.

2. VULNERABILITY NAME: Broken access

control SEVERITY: medium

PLUGIN: SSH weak HMAC Algorithms supported

PORT: N/A

DESCRIPTION: When access control checks are not applied consistently - or not at all - users are able to access data or perform actions that they should not be allowed to perform. This can lead to a wide range of problems, including information exposures, denial of service, and arbitrary code execution.

- : Ensure that the application has a robust authentication and authorization mechanism in place. Users should be authenticated before any sensitive actions are performed, and authorization checks should be conducted to verify that users have the necessary permissions for the requested operation.
- Ensure that session management is secure. Use strong session IDs, enforce session timeouts, and regenerate session tokens after login. Implement session fixation protection to prevent session hijacking.

BUSINESS IMPACT: wide-ranging business impact, including financial losses, reputation damage, legal consequences, operational disruption, and loss of customer trust. Addressing this vulnerability is crucial to protect the organization and its stakeholders from these potential negative effects.

3. VULNERABILITY NAME: SQL injection

SEVERITY: medium

PLUGIN: N/A (typically would be a security testing tool)

PORT: port 80/443

DESCRIPTION: The product constructs all or part of an SQL command using externally influenced input from an upstream component, but it does not neutralize or incorrectly neutralizes special elements that could modify the intended SQL command when it is sent to a downstream component.

SOLUTION:

- If parameterized statements are not feasible, use proper input validation and escaping mechanisms to sanitize user inputs before they are used in SQL queries. This helps prevent malicious code injection.
- Utilize stored procedures for database operations. This can help separate SQL code from application code and reduce the risk of SQL injection.
- Implement strict input validation for all user inputs, ensuring that data adheres to the expected format and structure.

BUSINESS IMPACT: In summary, it is crucial to underscore that CWE-89, known as SQL Injection, can exert a profound and diverse business impact. This encompasses critical facets such as data breaches, financial setbacks, harm to reputation, legal ramifications, and operational turmoil. Hence, the imperative of preventing and remedying SQL injection vulnerabilities cannot be overstated, as it is indispensable for fortifying the security and continuity of an organization's applications and data.

4. VULNERABILITY NAME: Broken

DESCRIPTION: The system's authorization functionality does not prevent one user from gaining access to another user's data or record by modifying the key value identifying the data.

SOLUTION:

- Implement strong authentication mechanisms, such as multi-factor authentication (MFA) or two-factor authentication (2FA), to ensure that only authorized users can access the system.
- Implement proper authorization controls to ensure that users have the necessary permissions to access specific functions or data. Use role-based access control (RBAC) to manage user privileges effectively.
- Deploy intrusion detection and prevention systems (IDS/IPS) to monitor and block suspicious activities related to authentication and authorization.

BUSINESS IMPACT: To effectively mitigate the business impact of CWE-285, it is imperative that organizations place a strong emphasis on bolstering their authentication and session management practices. This should encompass the adoption of multi-factor authentication, the secure storage of credentials, meticulous session handling, and a commitment to conducting routine security assessments and testing. The rectification of these vulnerabilities stands as a critical imperative, safeguarding sensitive data, user identities, and the organization's overarching security stature and reputation.

5. VULNERABILITY NAME: Insecure Direct object

Reference SEVERITY: Info

PLUGIN: Service Detection (plugin id:22964)

PORT: port 80/443

DESCRIPTION: Insecure Direct Object Reference (IDOR) is a vulnerability that arises when attackers can access or modify objects by manipulating identifiers used in a web application's URLs or parameters. It occurs due to missing access control checks, which fail to verify whether a user should be allowed to access specific data.

SOLUTION:

- Utilize parameterized queries or prepared statements when interacting with databases. This prevents user-controlled input from being executed as SQL code.
- Implement a Content Security Policy to mitigate the risk of cross-site scripting (XSS) attacks, which can be used to execute unauthorized queries.
- Implement account lockout mechanisms that temporarily suspend accounts after a specified number of unauthorized access attempts to prevent brute force attacks.

BUSINESS IMPACT: To effectively mitigate the business impact of CWE-639, organizations must make it a top priority to fortify their access control and authorization mechanisms. This entails the implementation of robust security measures, the regular conduct of comprehensive security assessments, and the deployment of intrusion detection systems to promptly identify and counter unauthorized access attempts. These measures stand as critical in the defence of sensitive data, the overall security of systems, and the preservation of the organization's esteemed reputation.

SEVERITY: low

PLUGIN: SSL cipher Block chaining cipher suites

PORT: port 443

DESCRIPTION: The product specifies permissions for a security-critical resource in a way that allows that resource to be read or modified by unintended actors.

SOLUTION:

- Implement a change management process to document and review all changes to system configurations. This helps prevent unauthorized or unintended alterations.
- Follow security guidelines provided by software and hardware vendors to ensure secure configurations.
- Regularly perform compliance checks against industry standards and regulations to ensure that configurations remain in compliance with security requirements.

BUSINESS IMPACT: To effectively lessen the business consequences stemming from CWE-732, organizations must place a strong emphasis on bolstering their configuration management practices. This involves the establishment of secure default settings, the routine conduct of comprehensive security evaluations, and the deployment of intrusion detection systems to promptly detect and address misconfigurations and vulnerabilities. These actions are of paramount importance in safeguarding sensitive data, fortifying system security, and preserving the organization's esteemed reputation.

6. **VULNERABILITY NAME:** Cross-Site Request

Forgery **SEVERITY:** Low

PLUGIN: HTTP Form-Based Authentication Detected

PORT: port 8080/443

DESCRIPTION: Cross-Site Request Forgery (CSRF) is an attack that forces an end user to execute unwanted actions on a web application in which they're currently authenticated. With a little help of social engineering (such as sending a link via email or chat), an attacker may trick the users of a web application into executing actions of the attacker's choosing. If the victim is a normal user, a successful CSRF attack can force the user to perform state changing requests like transferring funds, changing their email address, and so forth. If the victim is an administrative account, CSRF can compromise the entire web application.

SOLUTION:

- Implement Synchronizer Token Patterns to include unique, randomly generated tokens in each HTTP request and validate these tokens on the server side to verify the authenticity of the request.
- Utilize the Same Site attribute in cookies to restrict their usage to same-site requests only, preventing them from being sent along with cross-site requests and effectively mitigating CSRF. Implement robust encryption key management practices, including secure key storage, key rotation, and access controls.
- Utilize well-established and proven cryptographic libraries and algorithms for encryption and decryption

utilization of encryption and robust, secure protocols. The routine conduct of security assessments and testing is pivotal in pinpointing and remedying vulnerabilities linked to data transmission.

Furthermore, the education of users on secure data handling practices plays a vital role in proactively preventing data exposure incidents. These actions are of paramount importance in the protection of sensitive data and in preserving the trust of both customers and partners.

7. **VULNERABILITY NAME:** Clickjacking

- Utilize framework-specific protections and security features to prevent CSRF attacks. Many modern web frameworks have built-in mechanisms and libraries for handling CSRF vulnerabilities effectively.
- Implement a Content Security Policy to restrict the sources from which various types of content can be loaded. By specifying the trusted sources of content, you can minimize the risk of malicious code execution and reduce the likelihood of successful CSRF attacks.

BUSINESS IMPACT: To effectively reduce the business repercussions of CWE-352, organizations must prioritize secure coding practices, including the incorporation of anti-CSRF tokens, while also conducting routine security assessments and testing to detect and rectify CSRF vulnerabilities. Furthermore, educating users on secure browsing practices is pivotal in preventing CSRF attacks. These actions are of paramount importance in the protection of data and in upholding the trust of customers and partners.

8. VULNERABILITY NAME: Cleartext Transmission of Sensitive

Information SEVERITY: Medium

PLUGIN: Cleartext Transmission of Sensitive Information over HTTP

PORT: port 443

DESCRIPTION: The product transmits sensitive or security-critical data in cleartext in a communication channel that can be sniffed by unauthorized actors. Many communication channels can be "sniffed" (monitored) by adversaries during data transmission. For example, in networking, packets can traverse many intermediary nodes from the source to the destination, whether across the internet, an internal network, the cloud, etc. Some actors might have privileged access to a network interface or any link along the channel, such as a router, but they might not be authorized to collect the underlying data. As a result, network traffic could be sniffed by adversaries, spilling security-critical data.

SOLUTION:

- Educate users and employees on secure data handling practices, such as recognizing secure websites (look for "https") and avoiding insecure Wi-Fi networks.
- Employ data masking or redaction techniques to replace sensitive data with placeholders or cryptographic representations during transmission.
- Implement encryption for data in transit. Use secure encryption protocols such as TLS/SSL to protect sensitive information during transmission.
- Use secure communication protocols for transmitting data, such as HTTPS for web applications, and ensure that the selected protocols are kept up to date.

BUSINESS IMPACT: To effectively reduce the business consequences associated with CWE-319, organizations must prioritize the adoption of secure data transmission practices. This includes the

SEVERITY: Info

PLUGIN: HTML Rendering Layer

Vulnerability PORT: port 80/443

DESCRIPTION: The web application does not restrict or incorrectly restricts frame objects or UI layers that belong to another application or domain, which can lead to user confusion about which interface the user is interacting with.

SOLUTION:

- Set the X-Frame-Options HTTP response header to deny or same-origin to control how your site can be framed. This is supported by most modern browsers.
- Utilize frame-busting JavaScript code in web applications to prevent the embedding of your site within malicious frame's. This code can disrupt clickjacking attempts.
- Implement a Content Security Policy to restrict which domains can embed your site in iframes. This can help prevent unauthorized framing.
- Implement additional security controls to prevent UI redress attacks, such as clickjacking, within your web application.

BUSINESS IMPACT: To effectively reduce the business consequences resulting from CWE-1021, organizations must prioritize the implementation of protective measures, such as frame-busting code. Simultaneously, educating users on safe browsing practices plays a crucial role in preventing clickjacking incidents. Additionally, routine security assessments and testing are pivotal for identifying and mitigating vulnerabilities associated with clickjacking. These actions are of paramount importance in upholding user trust, ensuring data protection, and safeguarding the organization's reputation.

9. VULNERABILITY NAME: web server allows password auto-completion

SEVERITY: Low

PLUGIN: SSL/TLS Diffie-Hellman Modulus <= 1024 Bits (Logjam)

PORT: port 443

DESCRIPTION: Weaknesses in this category are related to the design and implementation of data confidentiality and integrity. Frequently these deal with the use of encoding techniques, encryption libraries, and hashing algorithms. The weaknesses in this category could lead to a degradation of the quality data if they are not addressed.

SOLUTION:

- Implement secure password storage mechanisms, such as using strong and salted cryptographic hashing algorithms like crypt or script.

- Use secure encryption protocols, like TLS/SSL, for data transmission over networks to protect data in transit

BUSINESS IMPACT: To effectively lessen the business consequences associated with CWE-310, organizations must prioritize the adoption of secure cryptographic practices. This entails ensuring proper password storage and robust encryption key management. Simultaneously, conducting routine security assessments and testing is pivotal in detecting and mitigating vulnerabilities related to cryptographic issues. Striving for compliance with pertinent data protection regulations and industry standards is equally essential. These actions are of paramount importance in the protection of sensitive data, the upholding of user trust, and the preservation of the organization's esteemed reputation.

S. NO	VULNERABILITY NAME	SEVERITY	PLUGINS
1.	Cross-site scripting (DOM based XSS)	Medium	Multiple XSS
2.	jQuery 1.3<3.5.0 Multiple XSS	Medium	CGI Abuse: XSS
3.	Information exposure (web server HTTP header internal IP disclosure)	Low	Webserver HTTP Header Internal IP Disclosure
4.	Information exposure (SSL (Multiple Issues))	Info	SSL Certificate Information
5.	Information exposure (HTTP (Multiple Issues))	Info	HTTP Information
6.	OS identification failed	Info	TCP/IP Remote Probing

TARGET WEBSITE: vit.ac.in

TARGET IP ADDRESS:

122.184.65.22

Report:

1. VULNERABILITY NAME: Cross-site scripting (DOM based

XSS) SEVERITY: Medium

PLUGIN: Multiple

XSS PORT: port 443

DESCRIPTION: It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a

user's browser. DOM-XSS: DOM stands for Document Object Model and is a programming interface for HTML and XML documents. It represents that programs can change the document structure, style and content. A webpage is a document, and this document can be either displayed in the browser window or as the HTML Source.

SOLUTION:

- The most important recommendation is to update to the latest version of jQuery. Newer versions typically include security patches, bug fixes, and performance improvements.
- Stay vigilant about jQuery updates and security announcements. Subscribe to relevant security mailing lists or forums to receive notifications about new releases.
- A Content Security Policy (CSP) can help mitigate various types of client-side vulnerabilities, including Cross-Site Scripting (XSS). Consider implementing a CSP in your web application to restrict the sources from which scripts can be loaded.

user trust, and safeguarding the organization's esteemed reputation.

2. VULNERABILITY NAME: Information Exposure (SSL (Multiple Issues)

) SEVERITY: Info

PLUGIN: SSL certificate

Information PORT: port 443

SOLUTION:

- When interacting with databases, use parameterized statements or prepared statements to avoid SQL injection, which can be a vector for XSS.
- Sanitize and validate all user inputs and ensure that any data displayed on the web page is properly encoded. Use output encoding libraries or functions to prevent script injection.
- Avoid using DOM manipulation methods that can introduce vulnerabilities. Be cautious with innerHTML, document.write, and other methods that can execute scripts.
- Consider using well-established JavaScript frameworks and libraries that include built-in security features to prevent DOM XSS.

BUSINESS IMPACT: The application stores dangerous data in a database, message forum, visitor log, or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area displayed to many users or particularly interesting users. Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data belonging to the user.

3. VULNERABILITY NAME: jQuery 1.2 < 3.5.0 Multiple XSS

SEVERITY: Medium

PLUGIN: CGI Abuse: XSS

PORT: port 443

DESCRIPTION: It occurs when a malicious script is injected directly into a vulnerable web application. Reflected XSS involves the reflection of a malicious script of a web application, onto a user's browser. Stored XSS (Persistent XSS): In this type of attack, the malicious script is permanently stored on a server or in a database, often in user-generated content like comments or forum posts. When other users view the affected content, the script executes in their browsers, potentially compromising their data or sessions. Reflected XSS: Reflected XSS occurs when the malicious script is embedded in a URL and immediately executed when a victim clicks on a manipulated link. The script is not stored on a server but is reflected off a web application, making it a one-time attack. DOM-based XSS: DOM stands for Document Object Model and is a programming interface for HTML and XML documents. It represents that programs can change the document structure, style and content. A webpage is a document, and this document can be either displayed in the browser window or as the HTML Source.

- If updating jQuery is not immediately possible, review your code for potential security issues. Ensure that you are escaping or sanitizing user inputs and that you are using safe coding practices to prevent XSS and other security vulnerabilities.

BUSINESS IMPACT: The application stores dangerous data in a database, message forum, visitor log, or other trusted data store. Later, the dangerous data is subsequently read back into the application and included in dynamic content. From an attacker's perspective, the optimal place to inject malicious content is in an area displayed to many users or particularly interesting users. Interesting users typically have elevated privileges in the application or interact with sensitive data that is valuable to the attacker. If one of these users executes malicious content, the attacker may be able to perform privileged operations on behalf of the user or gain access to sensitive data belonging to the user.

4. VULNERABILITY NAME: Information Exposure (Web Server HTTP Header Internal IP Disclosure)

SEVERITY: Low

PLUGIN: Web Server HTTP Header Internal IP Disclosure

PORT: port 80

DESCRIPTION: The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information.

SOLUTION:

- Carefully review your web server's configuration files (e.g., Apache's httpd.conf, Nginx's nginx.conf) to identify anywhere internal IP addresses are inadvertently included in HTTP response headers.
- Remove or replace any internal IP addresses in the server configuration files with appropriate placeholders or public IP addresses.
- If you use proxy servers or load balancers, ensure that they are configured to hide internal IP addresses and only expose public IP addresses instances in HTTP headers.
- Review and configure HTTP response headers (e.g., Server, X-Powered-By) to limit the exposure of server and technology details, reducing the potential for disclosing internal IP addresses.

BUSINESS IMPACT: To effectively reduce the business consequences arising from SSL issues, organizations must give utmost priority to establishing a secure SSL/TLS configuration, conducting frequent security assessments and testing. Adhering to data protection regulations and industry

DESCRIPTION: The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information.

SOLUTION:

- Configure your web server to use the latest and strongest encryption protocols, such as TLS 1.2 or higher. Disable outdated and insecure protocols like SSLv3.
- Implement Perfect Forward Secrecy to ensure that even if the private key is compromised, previously recorded encrypted traffic remains secure.
- Safeguard your private keys by storing them securely and ensuring they are not accessible to unauthorized parties.

BUSINESS IMPACT: To effectively reduce the business consequences arising from SSL issues, organizations must give utmost priority to establishing a secure SSL/TLS configuration, conducting frequent security assessments and testing. Adhering to data protection regulations and industry standards is imperative. These actions play a pivotal role in safeguarding sensitive data, upholding user trust, and safeguarding the organization's esteemed reputation.

5. VULNERABILITY NAME: Information Exposure (HTTP (Multiple

Issues)) SEVERITY: Info

PLUGIN: HTTP

information PORT: port 80

/443

DESCRIPTION: The product exposes sensitive information to an actor that is not explicitly authorized to have access to that information

SOLUTION:

- Minimize the use of unnecessary redirects as they can increase page load times. Use 301 (permanent) redirects for consistent content.
- Regularly check for broken links (404 errors) and fix them to ensure users find the content they are looking for.
- Properly configure CORS to control which domains are allowed to make requests to your server.
- Migrate your website from HTTP to HTTPS to ensure secure data transmission. This is essential for user trust and search engine ranking.

BUSINESS IMPACT: To effectively reduce the business repercussions of HTTP issues, organizations must place a strong emphasis on ongoing monitoring and enhancement of their web services. Consistent performance testing and security assessments, coupled with proactive maintenance, are indispensable to ensure a seamless and secure online presence. Furthermore,

offering timely and helpful customer support and transparent communication when HTTP issues arise is instrumental in mitigating adverse effects on user trust and satisfaction.

6. VULNERABILITY NAME: OS Identification

Failed SEVERITY: Info

PLUGIN: TCP/IP remote probing

PORT: N/A

DESCRIPTION: OS identification failed refers to the inability of a system or software to accurately recognize the operating system running on a device or network. This issue can result from various factors such as misconfigurations, network issues, or compatibility problems. The failure to identify the OS correctly can lead to security vulnerabilities, hindered software updates, and potential operational disruptions. Resolving this issue promptly is crucial to ensure accurate security measures and the smooth functioning of the system or network.

SOLUTION:

- Minimize the use of unnecessary redirects as they can increase page load times. Use 301 (permanent) redirects for consistent content.
- Regularly check for broken links (404 errors) and fix them to ensure users find the content they are looking for.
- Properly configure CORS to control which domains are allowed to make requests to your server.
- Migrate your website from HTTP to HTTPS to ensure secure data transmission. This is essential for user trust and search engine ranking.

BUSINESS IMPACT: It may lead to operational inefficiencies, potential security vulnerabilities, non-compliance with industry regulations, and a possible loss of customer trust. Such failures could result in downtime, compromised data security, regulatory penalties, and damage to the company's reputation, potentially leading to financial losses and decreased market competitiveness. Resolving this issue promptly is crucial to mitigate these risks and maintain the smooth functioning of the organization

Title:

Integrate Zscaler with Identity Providers (IdPs) such as Azure AD

Zero Trust Network Access (ZTNA) is a cybersecurity framework that shifts access control from a network-centric model to an identity-centric model. The foundational principle of ZTNA is "never trust, always verify," meaning that no user or device is trusted by default, regardless of whether they are connecting from inside or outside the corporate network. Unlike traditional remote access solutions that grant users broad access to the entire network, ZTNA creates a secure, application-specific tunnel between the user and the resources they are authorized to access. This ensures that the application remains invisible to unauthorized users and the broader internet, effectively shrinking the attack surface and eliminating the risk of lateral movement across the network.

Zscaler Private Access (ZPA)

Zscaler Private Access (ZPA) is a cloud-delivered ZTNA solution that provides identity- and context-based access to private applications without granting network-level access. ZPA operates on a "service-initiated" model, where users connect to the Zscaler cloud, and the cloud brokers a secure connection to the specific application via ZPA App Connectors deployed in the data center or cloud environment. Because ZPA never places the user on the corporate network, internal IP addresses and network infrastructure are completely hidden. Access policies are dynamically enforced based on user identity, group membership, device posture, and location, ensuring that each user sees only the applications they are explicitly permitted to use.

Principle of Least Privilege

The principle of least privilege (PoLP) is a security concept in which a user, device, or process is granted only the minimum levels of access—or permissions—necessary to perform its function. In the context of network security, this means moving away from "fat" network access (where authenticated users can reach large segments of the network) to "thin," application-specific access. By enforcing least privilege through ZPA, organizations ensure that even if a user's credentials are compromised, the attacker cannot pivot to other systems because the network itself is invisible and inaccessible. This principle is foundational to reducing insider threats and limiting blast radius during a security incident.

Identity-Aware Access

Identity-aware access is a security model that bases access decisions on who the user is, rather than where they are connecting from. ZPA integrates directly with existing identity providers (IdPs) such as Microsoft Active Directory, Azure AD, Okta, or Ping Identity. This integration allows security teams to leverage existing user attributes, group memberships, and roles to build granular access policies. For example, a policy can state that only members of the "Finance" group, using a company-managed device, can access the "SAP Finance" application during business hours. This ensures that access is tightly coupled with the user's identity and context, reducing reliance on static IP addresses or network segments.

Application-Specific Access Policies

Application-specific access policies are the core enforcement mechanism within ZPA. Instead of creating rules that allow traffic to entire subnets or IP ranges, administrators define policies that target individual applications or groups of applications. These policies are written in plain language logic and can incorporate multiple conditions, including user identity, device posture, location, and time of day. ZPA allows for the segmentation of internal applications—such as HR systems, internal wikis, development servers, and database interfaces—so that each application has its own unique access policy. This micro-segmentation ensures that a user authenticated for one application has no visibility or connectivity to any other application.

ZPA Architecture and Components

The ZPA architecture is composed of several key components that work together to provide secure, scalable access:

1. **Zscaler Cloud (Service Edge):** A globally distributed cloud platform that acts as the broker between users and applications. It handles authentication, policy enforcement, and connection brokering without sitting in the data path.
2. **App Connectors:** Lightweight software VMs or containers deployed within the private data center or cloud environments. These connectors initiate outbound connections to the Zscaler cloud, eliminating the need for inbound firewall rules. They establish secure tunnels to the ZPA edge and forward traffic to the internal applications.
3. **ZPA Client Connector:** A software client installed on end-user devices (Windows, macOS, Linux, iOS, Android). It authenticates the user, checks device compliance, and establishes a micro-tunnel to the ZPA edge for application access.
4. **Application Segments:** Logical groupings of applications defined within the ZPA admin console. These segments represent the private resources that users need to access.

Implementation Methodology

The deployment of ZPA follows a structured, phased approach to ensure minimal disruption and maximum security efficacy:

- **Phase 1: Discovery and Inventory:** Identify all internal applications currently in use. Categorize them by criticality, department, and user base. Tools like network scanners, configuration management databases (CMDB), and interviews with department heads are used to build a comprehensive application inventory.
- **Phase 2: Connector Deployment:** Deploy ZPA App Connectors in the data center or cloud environments where the applications reside. This involves provisioning virtual machines (using VMware, Hyper-V, or cloud-specific images) and configuring them to connect to the ZPA cloud.
- **Phase 3: Application Onboarding:** Define application segments within the ZPA console. This includes specifying the IP addresses and ports of the internal applications and associating them with the appropriate App Connector groups.
- **Phase 4: Policy Creation:** Integrate ZPA with the corporate identity provider (e.g., Azure AD). Create granular access policies based on user groups, device posture, and location. Policies are tested in a simulation mode before enforcement.

- **Phase 5: Pilot and Rollout:** Select a pilot group of users to test the new access method. Gather feedback, monitor performance, and refine policies. Gradually migrate user populations from legacy VPN access to ZPA.
- **Phase 6: VPN Decommissioning:** Once all users and applications are migrated, the legacy VPN infrastructure is decommissioned, eliminating a significant attack surface.

Tools and Technologies

- **Primary Platform:** Zscaler Private Access (ZPA) – The core Zero Trust access broker.
- **Identity Provider (IdP):** Microsoft Azure Active Directory (AD), Okta, or Ping Identity for user authentication and group synchronization.

Endpoint Protection Platform (EPP): Integration with endpoint security tools (e.g., CrowdStrike, SentinelOne, Microsoft Defender for Endpoint) to assess device posture and hygiene before granting

- access.
- **Security Information and Event Management (SIEM):** Integration with SIEM platforms (e.g., IBM QRadar, Splunk, LogRhythm) for centralized logging and correlation of access events.
- **Virtualization Platforms:** VMware vSphere, Microsoft Hyper-V, or public cloud environments (AWS, Azure, GCP) for hosting ZPA App Connectors.
- **Infrastructure as Code (IaC):** Tools like Terraform or Ansible for automated deployment and configuration of App Connectors at scale.
- **Digital Experience Monitoring (DEM):** Tools to monitor end-user connection quality and application performance.

Integration with Identity Providers

Integrating ZPA with the organization's existing identity provider is a critical step in establishing identity-aware access. This integration is typically done via SAML (Security Assertion Markup Language) or LDAP (Lightweight Directory Access Protocol). Once integrated, ZPA synchronizes user attributes and group memberships, allowing policies to reference these groups directly. For example, when a user authenticates via the ZPA Client Connector, they are redirected to the corporate IdP for

credentials and multi-factor authentication (MFA). Upon successful authentication, ZPA receives identity attributes that are then used to evaluate access policies and determine which applications the user is allowed to see and connect to.

Device Posture Checking

Beyond user identity, ZPA can enforce policies based on the security posture of the connecting device. Through integration with endpoint detection and response (EDR) and mobile device management (MDM) solutions, ZPA can assess factors such as:

- Antivirus software status and definition updates
- Disk encryption status (e.g., BitLocker, FileVault)
- Operating system patch level
- Presence of jailbreak or root compromise on mobile devices
- Real-time risk score from the EDR tool

If a device fails to meet the defined posture requirements (e.g., antivirus is disabled), ZPA can block access, grant limited access, or trigger a remediation workflow.

Micro-tunneling and Application Visibility

ZPA establishes a secure TLS/DTLS tunnel between the ZPA Client Connector and the ZPA service edge, and another tunnel between the service edge and the App Connector. Crucially, this creates a direct "app-to-user" micro-tunnel. The user's device is never placed on the corporate network; instead, traffic is proxied at Layer 7 (application layer). From the user's perspective, they can only resolve and connect to the specific DNS names or IP addresses of the applications they are authorized to access. All other internal resources remain completely invisible, preventing port scanning, vulnerability reconnaissance, and lateral movement.

Threat Intelligence Integration

To enhance security monitoring, ZPA can be integrated with threat intelligence platforms. Logs generated by ZPA—including connection attempts, policy violations, and user activity—can be forwarded to a SIEM (Security Information and Event Management) system. Within the SIEM, these logs can be correlated with threat intelligence feeds (such as those from MISP) to identify suspicious behavior. For example, if a user's credentials are used to access an application from an IP address associated with known malicious activity, the SIEM can trigger an alert, and integration with ZPA can automatically revoke that user's session.

Incident Response in a ZPA Environment

ZPA significantly streamlines the incident response process. When a potential threat is detected, security teams can leverage ZPA's capabilities to:

1. **Isolation:** Immediately revoke access for a compromised user account or device through the ZPA console, cutting off their connectivity to all internal applications.
2. **Forensics:** Access detailed logs within ZPA showing exactly which applications were accessed, by whom, from which location, and for how long.
3. **Containment:** Because users have no network-level access, a compromised endpoint cannot be used to scan or attack other internal systems, containing the blast radius.

4. **Granular Blocking:** Temporarily block access to specific high-risk applications while allowing access to others, ensuring business continuity during an investigation.

Understanding ZPA: Key Takeaways

Zscaler Private Access represents a fundamental shift in how remote access is architected and secured. Here are the key aspects to understand about the solution:

1. **It is cloud-native:** ZPA is delivered entirely from the Zscaler cloud, eliminating the need for hardware appliances and complex VPN head-end management.
2. **It is identity-centric:** Policies are built around users and groups, not IP addresses, making access control more intuitive and aligned with organizational structure.
3. **It ensures network invisibility:** By never placing users on the network, ZPA renders internal infrastructure invisible to the internet, preventing reconnaissance and DDoS attacks.
4. **It is scalable:** The cloud architecture allows ZPA to scale instantly to accommodate thousands of concurrent users without performance degradation.
5. **It supports hybrid environments:** ZPA works seamlessly for applications hosted in on-premises data centers, private clouds, and public cloud environments.

It improves user experience: Users get direct, optimized connections to applications without back-hauling traffic through a centralized VPN concentrator, reducing latency.

Conclusion:

Stage 1: In conclusion, web application testing is a crucial process that ensures the security, functionality, and reliability of web-based software. It involves a comprehensive evaluation of various aspects, including user interface, functionality, performance, and security to identify and rectify issues before they impact end-users. This process not only enhances the quality of web applications but also bolsters their resilience against potential threats and vulnerabilities. In an age where online services and applications are integral to our daily lives, effective web application testing is essential to deliver a seamless and secure user experience.

Stage 2: The Nessus report for the specific website has highlighted key security insights. It identified vulnerabilities, categorized by severity, with a focus on critical and high-risk issues that need immediate attention. Recommendations for addressing these vulnerabilities include patching, reconfiguration, and other security measures, with a priority on actions that can have the most significant impact. Compliance with industry standards and best practices is considered, ensuring alignment with recognized security benchmarks like OWASP guidelines. The report underscores the need for continuous monitoring, regular updates, and ongoing security efforts to maintain a strong security posture. In summary, the Nessus report offers a clear view of the website's security state, provides guidance on mitigating vulnerabilities, and stresses the importance of sustained vigilance against evolving threats.

Stage 3:

The implementation of Zscaler Private Access marks a pivotal shift from a vulnerable, network-centric security model to a resilient, identity-centric Zero Trust architecture. By enforcing least-privilege access and rendering private applications invisible to the internet, this project effectively neutralizes the threats of lateral movement and network-based attacks. The solution not only fortifies the organization's security posture against modern cyber threats but also delivers a seamless and high-performance experience for a distributed workforce. Furthermore, the decoupling of access from the network simplifies IT operations, reduces dependency on legacy VPN infrastructure, and provides granular visibility into user-application interactions. Ultimately, this initiative establishes a future-ready foundation for secure digital transformation..

Future Scope:

Stage 1: future scope of web application testing

The future scope of web application testing is promising and evolving rapidly, driven by the continuous growth of the internet and web-based applications.

Here are some key aspects of its future scope:

- **Increased Demand:** As more businesses and services move online, the demand for web applications continues to grow. This, in turn, increases the demand for web application testing to ensure security, reliability, and functionality.
- **Advanced Technologies:** The complexity of web applications is rising with the integration of advanced technologies like AI, IoT, and blockchain. Testing these cutting-edge applications will require specialized skills and tools.
- **Security Emphasis:** With the growing number of cyber threats and data breaches, security testing for web applications will become even more critical. Testers will need to focus on vulnerabilities, data protection, and compliance with regulations like GDPR.
- **Mobile and Cross-Browser Testing:** Mobile usage is on the rise, and web applications must be responsive and functional on various devices and browsers. Testing for cross-compatibility will be essential.
- **Performance Testing:** As user expectations for speed and performance increase, load testing, stress testing, and other performance testing types will be crucial to ensure web applications can handle traffic spikes.

Stage 2: future scope of the testing process you understood.

The future scope of the web application testing process, particularly in terms of identifying vulnerabilities, is exceedingly promising due to the continuous growth of digital platforms, the escalating importance of cybersecurity, and the ever-evolving threat landscape. Here are some key aspects that highlight the prospective scope of web application testing for vulnerability discovery:

- **AI-Driven Testing:** The integration of artificial intelligence (AI) and machine learning (ML) into web application testing tools will revolutionize the way vulnerabilities are identified. AI algorithms can swiftly scan vast amounts of code and data to pinpoint vulnerabilities that might elude human testers.
- **IoT and Mobile App Testing:** With the proliferation of Internet of Things (IoT) devices and the increasing usage of mobile applications, the scope of web application testing will extend to encompass these areas. Ensuring the security of IoT devices and the interfaces they use will become critical.
- **Blockchain and Cryptocurrency Security:** As blockchain technology and cryptocurrencies continue to gain traction, security testing will be imperative to safeguard transactions and digital assets. Ensuring the robustness of blockchain-based applications will be a specialized niche in web application testing.
- **Serverless Architectures:** With the adoption of serverless computing, web application testing will need to adapt to assess the security of serverless functions and APIs. Ensuring that serverless components are free from vulnerabilities will be paramount.
- **Container Security:** As containerization technologies like Docker and Kubernetes become ubiquitous, web application testing will extend to cover container security. This includes ensuring that container images and orchestration configurations are free from vulnerabilities.

Stage 3: future scope of Zero Trust

The successful deployment of ZPA lays the groundwork for a broader, more integrated security ecosystem. Future enhancements can be explored to further mature the organization's security capabilities:

- **Integration with Zscaler Internet Access (ZIA):** Converging Private Access (ZPA) with Internet Access (ZIA) to create a unified Secure Access Service Edge (SASE) platform, managing all web and private traffic through a single policy framework.
- **Advanced Device Posture Checking:** Moving beyond basic compliance to incorporate real-time risk scores from EDR tools, blocking access if a device shows signs of active compromise.
- **Privileged Remote Access:** Extending ZPA to third-party vendors and contractors with time-bound, audited access to specific critical systems, replacing jump boxes and bastion hosts.
- **IoT/OT Segmentation:** Applying the same micro-tunneling principles to Internet of Things (IoT) and Operational Technology (OT) devices to ensure they can only communicate with specific

management servers.

- **AI-Driven Anomaly Detection:** Leveraging machine learning on ZPA logs to detect behavioral anomalies, such as a user accessing an application at an unusual hour or from a strange location, triggering automated MFA challenges or access revocation.
- **Cloud Workload Protection:** Utilizing ZPA to secure connections between containers and microservices in multi-cloud environments (AWS, Azure, GCP), ensuring east-west traffic between workloads is encrypted and authenticated.
- **Automated Provisioning/Deprovisioning:** Enhancing the identity provider integration to automatically revoke application access the moment an employee is terminated or changes roles, ensuring near real-time access control.
- **Digital Experience Monitoring (DEM):** Implementing DEM tools to proactively monitor the end-user experience, identifying network latency or application performance issues before they impact productivity.

Topics Explored:

1. Zscaler
2. Zero trust
3. ZPA
4. VPN
5. MISP
6. College Network Information
7. Deployment of SOC in the College
8. Threat Intelligence
9. Incident Response

Tools Explored:

- Nessus Scanner